

Old Ideas, Novel Problems: The Instability of LLM-Based Novelty Evaluation

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Abstract

Automated ideation systems are often evaluated on the novelty of the ideas they produce, and that judgment is increasingly delegated to large language models. Such judges are typically built ad hoc and validated, if at all, on human-authored papers rather than on the generated ideas they are meant to score. So, how do novelty judges perform?

Not well. We present a systematic controlled study of novelty evaluation design choices. We first build an evaluation set automatically, mining OpenReview for passages where reviewers explicitly affirm or dispute a paper’s originality and keeping only submissions with unanimous agreement at the extremes of their research area; we pair these with ideas from a vanilla LLM generator. Across six judges, we find that small prompt design choices have large consequences; e.g., simply telling the judge that reviewers found one idea novel and the other not can change its verdict on more than half of the identical idea pairs it is shown, shifting pairwise accuracy by over 50 points and occasionally pushing it below chance. The same change helps one judge and hurts another. Retrieval and larger reasoning budgets help little, and two purpose-built novelty evaluators are outperformed by our cheapest prompted baseline. These results raise questions about reported novelty gains of automated ideation systems, and call for robust novelty evaluation methods.

1 Introduction

Recent years have seen a proliferation of systems that generate research ideas and propose new directions. A central criterion for judging such systems is *novelty*: whether the ideas they produce contribute a meaningful conceptual advance. Many works therefore automate this assessment, using it both to evaluate the ideas a system outputs and as an internal component of the ideation pipeline,

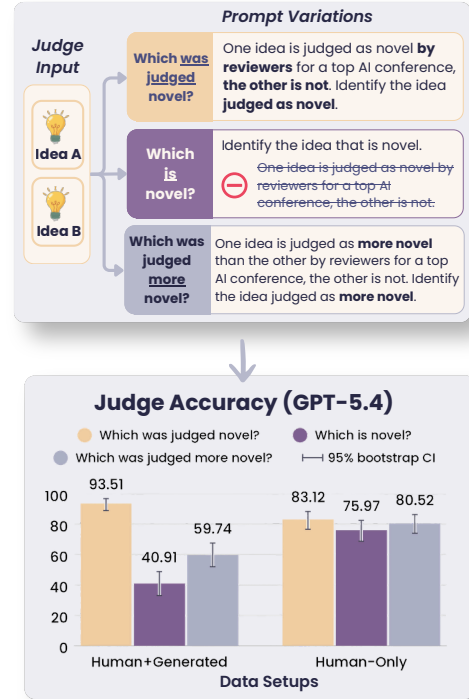


Figure 1: We run a **systematic controlled study** of novelty judges. Seemingly small changes, such as minor prompt variations, can wildly swing performance. Judges can grow markedly more *brittle* once LLM-generated ideas enter the data (Human+Generated), a failure that does not show up on the human-authored distribution they are typically tested on (Human-Only).

letting agents filter and refine a pool of candidate ideas (Wang et al., 2024; Yamada et al., 2025).

To date, there is no standard approach for novelty assessment, and papers suggesting a new ideation system typically ship their own ad hoc LLM judge (Radensky et al., 2026; Gottweis et al., 2025). Such judges are built around the system at hand, and might assume certain priors (e.g., a particular idea format). Their robustness, and the design choices behind them (such as their instructions, or whether they are allowed to access prior work), are rarely

tested or justified. Validation rests on small annotated sets (Radensky et al., 2026), outdated data that predates LLMs’ knowledge cutoffs (Shen et al., 2026), or human-authored papers (Si et al., 2024), a different distribution from the generated ideas the judge is actually applied to (Chen et al., 2026).

We present a systematic, extensive study of LLM-based novelty evaluation. Through controlled experiments, we vary different design choices practitioners make: which LLM is used as a judge and how the judge is prompted, whether evaluation is *pointwise* (is this idea novel?) or *pairwise* (which of these two is more novel?), its reasoning budget, whether it may retrieve related work, and the format of the judged ideas. We evaluate judges on human-authored ideas, and explore how their behavior changes when human-authored and LLM-generated ideas are judged together. Additionally, we benchmark off-the-shelf LLMs against specialized novelty judges (Yamada et al., 2025; Shahid et al., 2025).

To enable our experiments, we construct a benchmark using an automated pipeline that mines high-precision novelty labels from peer reviews (Figure 2). Unlike prior work that relies on coarse proxies such as paper acceptance (Qiao et al., 2026) or outdated novelty scores that venues no longer publish (Schopf and Färber, 2026), we derive labels from the underlying review text. Specifically, we isolate passages where reviewers explicitly affirm or dispute the originality of a contribution, retaining only submissions with unanimous consensus that fall within the top or bottom decile of their venue. We complement these with LLM-generated ideas, sampled under settings that with high probability yield lower novelty than the top human-authored ideas.

Using this benchmark, we find that novelty judges are strikingly brittle. Small changes to the evaluation setup, such as telling the judge that reviewers found one of two ideas novel, change a judge’s verdict on more than half of the identical pairs it is shown and swing its accuracy by more than 50 points (Figure 1). Judges also behave differently on AI-generated ideas than on human-authored ones, exposing a systematic gap in current evaluation practice: pairwise judges degrade once the comparison involves generated ideas, and grow markedly more brittle there. No single fix helps across the board. Judges are calibrated differently from one another, and the techniques commonly reached for to improve them, such as reasoning ef-

fort and retrieval, have limited effect and can even degrade some models. We further show that the additional complexity and monetary investment in dedicated novelty-evaluation models do not necessarily translate to better novelty verdicts.

Taken together, our results show that LLM novelty judges are highly unstable, **calling into question the robustness of a large body of prior work in which LLM judges were used to evaluate the novelty of AI-generated scientific ideas.**

Our contributions are as follows:

- A systematic controlled study of novelty judges, spanning six judge backbones, pointwise and pairwise judgments, and common design choices such as prompt wording, retrieval, reasoning effort, and idea format. Additionally, we compare judge behavior on human-written and LLM-generated ideas.
- An automated, reusable pipeline that mines explicit novelty signals from peer reviews to build novelty evaluation benchmarks. Using this pipeline, we construct two core evaluation setups: Human-Only (strongest high-novelty vs. weakest low-novelty human papers) and Human+Generated (strongest high-novelty vs. ideas from a deliberately simple LLM generator). The pipeline can be rerun on new conference data, and we release all data and code as an open resource.
- An empirical finding that judge verdicts depend heavily on how the evaluation is configured: the same judge returns different verdicts on the same instances under different configurations. This effect can be more pronounced on generated ideas, the regime where these judges are actually used, and common fixes (retrieval, increased reasoning) offer limited help. Spending more does not always buy better performance on this task.

2 Related Work

Automated research ideation. Systems that propose research ideas have proliferated in recent years, spanning end-to-end agentic scientists (Lu et al., 2026; Yamada et al., 2025), multi-agent pipelines that debate and refine candidate hypotheses (Gottweis et al., 2025; Su et al., 2025), and literature-grounded ideators that search over or recombine prior work (Wang et al., 2024; Hu et al.,

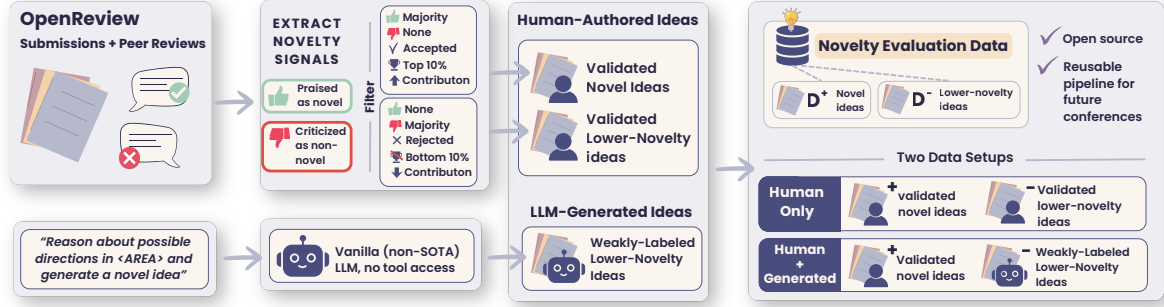


Figure 2: **Automatic data collection process.** We mine novelty labels from OpenReview, keeping only submissions whose reviewers agree on their originality, and complement them with weakly labeled lower-novelty ideas from a non-SOTA LLM ideator. The result is an open-source novelty evaluation dataset with two setups, Human-Only and Human+Generated. We instantiate the pipeline on ICLR 2026, but it can be rerun on future conferences.

2024; Li et al., 2024; Baek et al., 2025; Radensky et al., 2026).

Novelty evaluation is an important metric across these designs, yet there is no standard benchmark, evaluation set, or metric that prior work uses to measure it. Each work instead ships its own novelty judge (Si et al., 2024; Radensky et al., 2026), assembled from ad hoc design choices (such as custom prompts, backbone models, and prior-work access) without justification and without accounting for bias. This makes the reliability of the findings questionable, and complicates comparison between different ideation systems.

Our work targets this gap by performing the first systematic study of automated novelty evaluation. We examine key design choices and report their effect across various judges and ideation models.

Automatic novelty evaluation. Large-scale evaluations of novelty judges have so far used human-authored ideas. Examples are Moussa et al. (2026); Wu et al. (2026), who align judges with peer-review content, Schopf and Färber (2026); Qiao et al. (2026), who extract novelty labels from peer-reviewed papers, and Lin et al. (2024), who use a temporal proxy that treats the more recent of two arXiv papers published years apart as the more novel. Liu and Zhai (2026) forgo labels altogether, testing instead whether a novelty metric’s score moves in the expected direction when the pool of prior work is perturbed, for example by inserting the paper itself into the pool or by removing the papers it cites.

While some prior work does examine generated ideas, it rests on human annotation and is consequently small (tens of ideas) and highly specific.

Si et al. (2024) present a small dataset of mostly student-annotated ideas, focusing on a single generation system and 7 LLM-related research topics. Shahid et al. (2025) evaluate a literature-grounded novelty checker on a small set of ideas produced by one system (Radensky et al., 2026), and Gupta and Pruthi (2025) have annotators trace generated research documents back to prior work, targeting plagiarism rather than novelty. To our knowledge, no prior work systematically studies automatic novelty evaluation of AI-generated ideas.

Robustness of LLM judges. LLM judges are known to shift their verdicts with presentation order (Wang et al., 2023), with the choice between pointwise and pairwise formats (Liusie et al., 2024), and with surface wording alone (Raina et al., 2024; Du et al., 2026; Thakur et al., 2026). Such findings have seen little examination in the context of novelty judgment. We evaluate them in that setting and report the resulting variation.

3 An Automated Pipeline for Creating Novelty Evaluation Benchmarks

At a high level, our goal is to evaluate novelty judges in a range of different settings: ideas spanning higher and lower levels of novelty, authored both by humans and by automated ideation systems. We collect ideas and organize them into two conceptual pools, such that ideas in one are expected to be more novel than those in the other. These pools support both pointwise judgments of whether an individual idea is novel and pairwise judgments of which of two ideas is more novel. This section describes how we automatically collect ideas and their corresponding novelty labels

from multiple sources, and how we assemble them into our benchmark. Our collection pipeline is automated, allowing the benchmark to scale to larger idea pools. It is also reusable: it can be re-run on the reviews of future conference cycles, keeping the benchmark evergreen. Notably, our pipeline is not tied to ICLR: it applies to any venue that publishes free-text reviews, such as NeurIPS, which releases reviews for accepted papers and for rejected ones whose authors opt in, or ACL Rolling Review¹. The automatic data collection process is depicted in Figure 2.

3.1 Labeled Ideas Collection

Validated novel ideas. We extract novelty labels from ICLR 2026 submissions and their corresponding reviews. While some prior work builds idea-assessment datasets from review data, it derives either no explicit quality labels (Moussa et al., 2026; Wu et al., 2026), or labels from coarse proxies such as acceptance decisions (Qiao et al., 2026). Such proxies fold additional quality dimensions into the verdict, and a paper can be accepted while its reviewers consider the contribution incremental. A third line relies on novelty scores that venues no longer publish (Schopf and Färber, 2026), so the procedure cannot be re-run on newer submissions.

Reviewers do state verdicts on originality, but in prose, scattered across the free-form strengths and weaknesses sections. We therefore derive our labels from the review text itself: we prompt `claude-opus-4-6` to extract *novelty signals*, snippets of text where reviewers affirm or dispute the novelty of the *contribution itself*. Appendix A.1 lists the extraction prompt used and signal examples.

In our setting, an idea is deemed novel when its reviewers judge the contribution original and the submission sits at the top of the venue. The first is established by the novelty signals themselves: a majority of reviewers give a positive novelty signal, and no reviewer gives a negative one. The second is established by the review scores: the paper is accepted, its average rating falls in the top decile of its ICLR primary area, and its average contribution score clears a fixed floor. We treat each submission’s abstract as its idea.

Validated lower-novelty ideas. The complementary pool inverts every criterion. Its reviewers must flag issues with the contribution’s originality: a

	Ideas		Eval. instances	
	D_+	D_-	Pairwise	Pointwise
Human-Only	154	145 ICLR	154	299
Human+Generated	154	154 LLM	154	308

Table 1: **Data setups statistics.** D_+ holds the novel ideas and D_- the lower-novelty ones. Both setups share D_+ (154 ICLR-validated novel ideas) and differ in D_- : validated lower-novelty ICLR ideas in Human-Only, and LLM-generated ideas in Human+Generated.

majority give negative novelty signals, and no reviewer gives a positive one. Its submission must sit at the bottom of the venue: rejected, with an average rating in the bottom decile of its ICLR primary area and an average contribution score below a fixed ceiling. Appendix A.1 lists additional details on mining novelty labels from human reviews.

Weakly labeled lower-novelty ideas. Work that evaluates novelty judges at scale does so exclusively on *human-authored* papers (Schopf and Färber, 2026; Moussa et al., 2026), rather than on the machine-generated ideas that are the target distribution when evaluating ideation systems. Conversely, work that does examine generated ideas relies on expert annotation (Si et al., 2024; Shahid et al., 2025) and is consequently small (tens of ideas). Such annotations are slow and expensive to collect, and, once released, liable to leak into the training data of the very models later used as novelty judges. Worse, these expert labels are themselves unstable, as shown by Si et al. (2025), where the scores of LLM-generated ideas, including their novelty scores, dropped significantly once the ideas were executed and reviewed again.

We therefore build the generated pool under an explicit weak-labeling assumption: *every idea in the generated pool is less novel than every validated novel idea*. We design the pool to keep this assumption conservative. Its human side is the *validated novel* pool described above: top-decile accepted ICLR papers whose reviewers, having read the full paper, unanimously affirmed the novelty of the contribution. Its generated side comes from `claude-sonnet-4-5`, a deliberately non-frontier model released over a year ago, which is given only the ICLR primary area of its human counterpart and asked to “generate a novel idea” in a single pass, with no literature access, tools, scaffold, or feedback. We argue this is highly plausible given

¹<https://arr-data.aclweb.org/resources/>

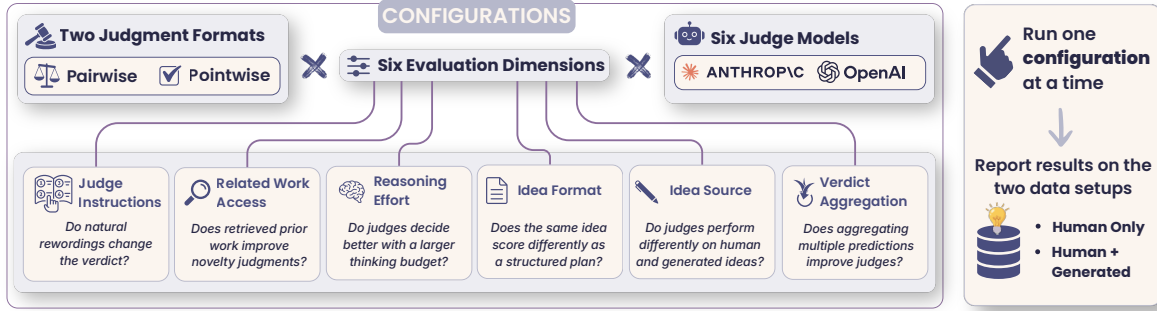


Figure 3: **Study overview.** We cross two judgment formats with six judge backbones and six evaluation dimensions. We run one configuration at a time and report the results on both data setups, Human-Only and Human+Generated.

the state of technology in 2025².

As a check that the assumption is not obviously violated, a computer science professor with broad expertise across AI areas judged 30 pairs from Human+Generated blind to source and with idea order shuffled, giving written justifications and citing prior work where relevant. To stress the assumption where it is most likely to fail, 28 of the pairs were ones our strongest pairwise judge (claude-opus-4-6) got wrong under the reference configuration, and two were controls it got right. The expert sided with the human idea in all 30 pairs, with no ties, over roughly five hours of annotation. We present this as a sanity check on the hardest cases rather than a validation of every label (see Appendix A.3 for details and examples). We also normalize the style of human and generated ideas to evaluate potential judge shortcuts (Section 5).

Finally, this labeling assumption has no bearing on our evaluation of *instability*: two configurations that return different verdicts on the same pair cannot both be right, whatever its label.

3.2 Benchmark Construction

Pointwise vs. pairwise evaluation. We consider the two forms of novelty evaluation in common use: (1) *Pointwise* (Shahid et al., 2025; Yamada et al., 2025), where the judge is given a single idea and decides whether or not it is novel (binary classification), and (2) *Pairwise* (Qiao et al., 2026), where the judge is given two ideas and decides which of the two is more novel (ranking). Beyond their prevalence, pointwise and pairwise evaluations are the atomic components underlying more complex assessment protocols, such as majority voting and tournament-style ranking (Si et al., 2024; Gottweis

et al., 2025).

Data setups. We assemble the collected ideas into two setups, each a pair of pools: D_+ (*novel ideas*), and D_- (*lower-novelty ideas*), populated so that an idea drawn from D_+ can be assumed more novel than one drawn from D_- . In Human-Only, D_+ holds the validated novel ideas and D_- the validated lower-novelty ideas, so both sides are human-authored. In Human+Generated, D_+ is unchanged and D_- holds the weakly labeled generated ideas. The two setups thus share D_+ and differ only in the source of D_- . Accordingly, D_- does not denote a fixed set of ideas, but whichever pool fills the lower-novelty side of the setup at hand.

Each data setup supports both pointwise and pairwise judgment, and both draw on both pools. In *pointwise*, every idea in either pool is one instance, labeled by its pool: ideas in D_+ are labeled novel and ideas in D_- not novel. In *pairwise*, each idea in D_+ is matched with one from D_- in the same ICLR primary area (e.g., *generative models*), so the two sides are topically comparable. In Human-Only, where D_- is the smaller pool, an idea may be matched more than once. Table 1 reports the resulting statistics.

Idea format. An idea in our benchmark is a single abstract, human-authored or generated. A submission’s own abstract advertises experimental outcomes that an LLM ideator, with no execution, cannot produce without hallucinating results. We therefore apply claude-opus-4-6 to every human abstract to remove evaluation artifacts (e.g., numerical results, claims of superiority to specific baselines). Prompt 2 in Appendix A.1 presents the full instructions for this step. Generated ideas are emitted in this same format and need no such treatment.

²See discussion on using newer models in Appendix C.2.1.

4 Experimental Setup

We perform a systematic study of novelty judges, spanning both judgment formats, a range of judge backbones, and several evaluation dimensions, on both data setups (Figure 3). Each run modifies a *single* design choice relative to a *reference configuration*, holding the rest of the evaluation process fixed. Table 2 summarizes the reference and the controlled changes we apply to it. Additional implementation details are in Appendix C.1.

4.1 Reference Configuration

Given a single idea (pointwise) or a pair of ideas (pairwise), we ask the judge to return a novelty verdict together with the reasoning behind it. Judges are prompted with `reasoning_effort = high` (when they expose this parameter) and with no access to literature search.

To improve prediction stability, we aggregate multiple verdicts from the judge. In the pointwise case we take the majority vote of three calls. In the pairwise case we call the judge three times per presentation order (A vs. B and B vs. A) to mitigate position bias (Wang et al., 2023). We score each idea by the fraction of the six calls that select it and award the comparison to the idea with the higher score. When the two ideas receive equal scores, we record a **tie**, i.e., the judge is unable to distinguish between them. The exact reference prompts and implementation details are given in Appendix B.1.

Both data setups share the same D_+ , the 154 validated novel ICLR ideas, and differ only in D_- (Section 3.2, Table 1). In Human-Only, D_- holds the validated lower-novelty ICLR ideas. In Human+Generated, D_- is populated by a non-SOTA model (claude-sonnet-4-5) prompted to “generate an idea” with no scaffold, literature access, or tools (Section 3.1). Since the generator fixes the low-novelty side of the benchmark, we also treat it as one of the controllable parameters in the study.

4.2 Explored Evaluation Dimensions

Judge prompt. The instructions given to the judge. We explore six variants: the reference prompt and five modifications (P1–P5, specified in Table 2). P1 and P2 weaken the novelty definition given in the reference prompt: P1 drops the instruction that warns against trivial or incremental contributions, and P2 removes the definition of

novelty altogether. P3–P5 (a family of pairwise prompts inspired by Si et al. (2024)) simply ask the judge to choose between two ideas, differing only in small changes: P3 tells the judge that reviewers at a top AI conference found one idea novel and the other not, P4 omits this reviewer framing and directly asks which idea is novel, and P5 retains the reviewer framing but asks which idea was judged *more* novel.

P3–P5 introduce small, local changes to the instructions: whether the judgment is framed through prior peer review and whether novelty is expressed in binary or comparative terms. Will these choices affect how the judge interprets novelty? Our question is how strongly the resulting evaluations depend on such seemingly minor prompt-design decisions, which practitioners must make when constructing a novelty judge. These prompts are not semantically identical; nevertheless, each asks the judge to select between the same two ideas on the basis of novelty. They represent closely related formulations that a practitioner could reasonably consider when implementing the same evaluation goal (indeed, the prompts closely follow Si et al. (2024)). Some variation in judgments is therefore unsurprising; our question is how large that variation becomes. We find that these local changes can produce substantially different verdicts and performance estimates, particularly when evaluating generated ideas.

Retrieval. Whether the judge sees related work alongside the idea. In the retrieval condition the judge additionally receives 5 abstracts of related work, collected with a Paper-Finder³ pipeline: we first isolate the contributions of the judged idea, then generate 5 targeted queries that probe the novelty of those contributions, and finally keep, for each query, the most relevant paper published before a fixed retrieval cutoff (2025-03-01). The cutoff sits about six months before the ICLR 2026 abstract deadline (2025-09-19). Authors often post drafts to arXiv well before submitting, and this margin keeps such preprints, which would reveal the very idea under evaluation, out of the retrieved work. We also adjust the prompt accordingly, asking the judge to assess novelty with respect to the retrieved work.

Reasoning effort. The inference-time reasoning budget given to the judge. Assessing novelty

³<https://github.com/allenai/asta-paper-finder>

Change	Description	Protocol		Data setup		Fig.
		Pt	Pw	HO	HG	
Judge prompt (<i>reference: requests judges to assess ideas against a novelty criterion defining what counts as novel</i>)						
P1: Criteria without guardrails	Weaker criterion, w/o guardrails (e.g., no trivial combinations)	✓	✓	✓	✓	4
P2: Criteria removed	Drop novelty criterion entirely	✓	✓	✓	✓	4
P3: “Which was judged novel?”	Prior review found exactly one idea novel; judge predicts which	–	✓	✓	✓	4b
P4: “Which is novel?”	P3 minus the presumption of prior review	–	✓	✓	✓	4b
P5: “Which was judged more novel?”	Prior review found one idea <i>more</i> novel; judge predicts which	–	✓	✓	✓	4b
Evaluation setup (<i>reference: no retrieval; high reasoning effort; free-text abstracts; verdict aggregation; D_– by sonnet-4-5</i>)						
+ Related-work (retrieval)	Judges receive 5 abstracts of related work	✓	✓	✓	✓	4
Reasoning effort = low	Reduced reasoning budget	✓	✓	✓	✓	4
Idea format	A two-field structured plan (Purpose Mechanism)	✓	✓	✓	✓	4
No verdict aggregation	Query pointwise judges once, pairwise once per direction	✓	✓	✓	✓	4
D _– source	D _– regenerated by opus-4-5, gpt-5.1, gpt-5.4; D ₊ fixed	✓	✓	–	✓	13
Judge backbones						
gpt-5.1, gpt-5.2, gpt-5.4, claude-sonnet-4-5, claude-opus-4-5, claude-opus-4-6						

Table 2: **Controlled changes.** Each change modifies a single design choice relative to the reference configuration, holding the rest of the novelty evaluation process fixed. Columns mark the protocols (Pt: pointwise, Pw: pairwise) and data setups (HO: Human-Only, HG: Human+Generated) each change is run under. Beyond these single changes, we compare prompted judges to dedicated novelty judges (Fig. 6) and to a strong agentic RAG judge (Fig. 5).

requires comparing an idea against the judge’s knowledge and deciding whether the overlap with prior work is substantial, which we expect to be reasoning-intensive. We test that expectation by lowering `reasoning_effort` from *high* to *low* for the backbones that expose the parameter.

Idea format. The surface form in which ideas are presented to the judge. Prior work presents ideas in different formats, from free-form abstracts (Wang et al., 2024) to structured research plans (Shahid et al., 2025; Si et al., 2024), typically without justifying the choice. We therefore test the effect of converting every idea into a two-field *purpose / mechanism* plan, which holds the content of the idea fixed and changes only its presentation.

Verdict aggregation. The number of judge calls we pool into a single decision. Aggregation is a simple but costly way to stabilize verdicts (Wang et al., 2023). We test whether dividing the number of aggregated calls by 3 (calling pointwise judges once, pairwise judges once per presentation order) degrades the judges.

D_– source. The ideation model that produces the lower-novelty pool. We hold D₊ fixed and regenerate D_– with three further backbones (claude-opus-4-5, gpt-5.1, gpt-5.4), covering a weaker and a stronger model from each of two vendors.

Judge backbone. We run our controlled study with a range of judge backbone models (gpt-5.{1,2,4}, claude-sonnet-4-5, claude-opus-4-{5,6}), all with training cutoffs predating the ICLR 2026 submission deadline so that the evaluated ideas fall outside their training data. The novelty labels are further still out of reach: they are derived from reviews that were only released on OpenReview on 2025-11-12, roughly two months after the deadline. Thus, even if an idea reached a judge’s training data through an earlier preprint, the reviewers’ verdict on its novelty could not have.

4.3 Dedicated Novelty Judges

Our study focuses on prompted novelty judges: off-the-shelf LLMs instructed to return a novelty verdict. This is the setting in which novelty is assessed in most ideation systems (Li et al., 2024; Hu et al., 2024; Shen et al., 2026). However, we also perform a focused comparison with the more complex evaluation pipelines presented in prior work. Specifically, we examine two such judges. The first is **Idea Novelty Checker** (Shahid et al., 2025): a literature-grounded pipeline that retrieves a broad candidate pool by keyword search, narrows it through embedding-based filtering and LLM re-ranking, and assesses novelty against the surviving papers, guided by expert-labeled examples. The second is **AI-Scientist** (Yamada et al., 2025): a

ReAct-style agent with Semantic Scholar access. The agent operates in a loop, where on each iteration it issues a search query, retrieves papers, and either decides on the idea’s novelty based on the results or continues searching.

4.4 Evaluation Metrics

For pointwise evaluation we report **F1** per class (novel / not-novel) and a macro average. For pairwise, we report two versions of accuracy, differing in how they treat ties: (1) **soft-accuracy**: ties receive a credit of 0.5; (2) **strict-accuracy**: ties receive no credit at all.

We assess **statistical significance** with a two-sided paired non-parametric bootstrap (100,000 resamples) over the per-metric difference between each change and the reference configuration, using BCa 95% confidence intervals. * marks differences whose 95% CI excludes 0 (i.e., $p < 0.05$).

5 Results

Figure 4 reports the ablation results for both pointwise (Figure 4a) and pairwise (Figure 4b) judgments. The first thing to note is that judges generally underperform, and rarely approach perfect results on this task. This happens even in the Human-Only setting, where D_+ and D_- are drawn from opposite ends of the ICLR rating distribution and should be maximally separable.

Performance is also highly unstable under configuration changes. For instance, removing the guardrails from the novelty definition, or dropping the definition from the prompt entirely (P1–P2), generally degrades pairwise judges while helping some pointwise ones. On pairwise Human+Generated, replacing the reference prompt with one that asks which idea human reviewers judged as novel (P3) improves every judge (by more than 20 points in some cases). Yet a variant of that same prompt (P4), asking which idea is novel without referring to any prior human judgment, makes gpt-5.4 collapse below chance level. Because both prompts judge the same pairs, the 52.6-point gap between P3 and P4 means the two prompts return different verdicts on at least 52.6% of the pairs; whichever prompt is right on a pair, the other is wrong, so this instability does not depend on our labels.

To rule out the possibility that judges are reacting to stylistic differences between human-written and LLM-generated ideas rather than to their nov-

elty, we rewrite all ideas into a shared plan format and rerun a subset of the changes that produced the largest shifts. The direction of the shifts is unchanged, and instability even grows: some changes flip the verdicts on more than two thirds of the examples (Figure 12, Appendix C.2).

These shifts also do not move models together. The same change might be neutral or even beneficial for one judge, but destructive for another. This is seen most clearly in pointwise judging (e.g., P1–P2), but also in pairwise judgments: retrieval is a large gain for claude-opus-4-5 on Human+Generated while neutral for others. A configuration validated on one judge therefore carries no guarantee for the next.

We notice that design choices prior work commonly reaches for, like increased reasoning (Saha et al., 2025) or retrieval (Shahid et al., 2025; Lin et al., 2024), have a limited and unexpected effect. Lowering the reasoning effort from "high" to "low" costs at most 7 performance points, and is usually neutral. Surprisingly, some models even benefit from using a smaller reasoning budget. Retrieval is similarly underwhelming: it induces a statistically significant improvement of at most 16 points (less than replacing the reference prompt with P3), only for some models, and never brings the absolute numbers near 100. This remains true when D_- is generated by other ideation backbones, where retrieval even significantly degrades performance in some cases (Figure 14).

The effect of the same prompt change also depends strongly on the evaluated data. For gpt-5.4, switching from P3 to P4 changes strict accuracy by approximately 7 points on Human-Only, compared with 52.6 points on Human+Generated. Thus, a prompt choice that appears relatively benign when validating on human-authored ideas can have much larger consequences when the judge evaluates generated ideas, highlighting the importance of validation on the intended distribution.

Interestingly, removing instructions against trivial or additive novelty from the prompt (P1) harms every judge once AI-generated ideas are involved, while being mostly neutral on human-authored pairs. This might indicate that LLM-generated ideas lean more towards this style of ideation.

We also vary the ideation model that generates D_- , and find that judges generally degrade once the lower-novelty ideas are LLM-generated rather than human-authored, with pairwise accuracy falling below chance against some backbones (Figure 13,

		Human-only						Human + generated					
Change	Reference	60.7	70.7	80.5	75.2	64.4	66.6	69.1	78.4	88.9	75.0	64.5	70.5
	P1 · Criteria without guardrails	+15.7* (76.4)	+1.7 (72.4)	-6.7* (73.9)	-6.0* (69.2)	+7.1* (71.6)	+10.0* (76.6)	+12.7* (81.8)	-8.2* (70.3)	+0.4 (89.3)	-5.7* (69.3)	+6.3* (70.8)	+6.9* (77.5)
	P2 · Criteria removed	+10.3* (71.0)	+6.0* (76.7)	+0.7 (81.2)	-1.2 (74.0)	-5.9* (58.5)	-5.2* (61.4)	+8.9* (78.0)	+3.9 (82.4)	+3.6* (92.5)	+0.9 (75.9)	-4.9* (59.6)	-5.9* (64.6)
	+ Related work (retrieval)	+7.9* (68.6)	+5.8* (76.4)	-0.3 (80.2)	-2.4 (72.8)	+2.3 (66.7)	+6.0* (72.6)	-1.5 (67.6)	-1.8 (76.6)	+0.3 (89.2)	-2.0 (73.0)	+5.2* (69.7)	+1.9 (72.5)
	Reasoning effort = low	—	+2.7 (73.4)	-6.7* (73.9)	-4.9* (70.3)	-0.4 (64.1)	+6.0* (72.6)	—	+2.5 (81.0)	-3.4* (85.4)	+1.2 (76.2)	+3.2 (67.7)	+6.1* (76.6)
	No verdict aggregation	-0.6 (60.2)	+1.4 (72.1)	-2.3* (78.2)	-3.0 (72.2)	-4.7* (59.7)	-1.1 (65.5)	-2.7* (66.5)	-1.1 (77.4)	+1.3 (90.2)	+2.0 (77.0)	+1.1 (65.6)	+1.3 (71.9)
	Idea as research plan	+0.8 (61.5)	+1.2 (71.9)	-2.2 (78.4)	+1.4 (76.6)	-6.3* (58.2)	-3.1 (63.5)	-3.5 (65.6)	-0.4 (78.1)	+0.1 (88.9)	+1.6 (76.6)	-6.3* (58.2)	-8.2* (62.4)
			sonnet-4-5	opus-4-5	opus-4-6	gpt-5.1	gpt-5.2	gpt-5.4	sonnet-4-5	opus-4-5	opus-4-6	gpt-5.1	gpt-5.2
		Judge model											

* p < 0.05 (two-sided paired bootstrap)

* p < 0.05 (two-sided paired bootstrap)

(a) Pointwise judging, macro-F1.

		Human-only						Human + generated							
Change	Judge prompt	Reference	74.0	79.2	82.5	83.8	83.8	83.1		39.6	63.0	86.4	63.0	76.0	77.9
		P1 · Criteria without guardrails	-9.7* (64.3)	+0.6 (79.9)	-1.9 (80.5)	+0.6 (84.4)	-1.3 (82.5)	-2.0 (81.2)		-21.4* (18.2)	-14.9* (48.0)	-5.8* (80.5)	-9.7* (53.2)	-11.7* (64.3)	-9.7* (68.2)
		P2 · Criteria removed	-9.7* (64.3)	-0.7 (78.6)	-1.3 (81.2)	+0.6 (84.4)	+0.0 (83.8)	-0.7 (82.5)		-16.2* (23.4)	-5.8 (57.1)	-3.9* (82.5)	-4.5 (58.4)	-0.7 (75.3)	+1.9 (79.9)
		P3 · "Which was judged novel?"	+0.7 (74.7)	-3.9 (75.3)	-3.9 (78.6)	+2.6 (86.4)	+0.0 (83.8)	+0.0 (83.1)		+20.8* (60.4)	+29.2* (92.2)	+13.6* (100.0)	+16.2* (79.2)	+5.8* (81.8)	+15.6* (93.5)
		P4 · "Which is novel?"	-4.5 (69.5)	-11.7* (67.5)	-24.7* (57.8)	-11.0* (72.7)	-4.5 (79.2)	-7.2* (76.0)		-3.3 (36.4)	-4.5 (58.4)	-13.6* (72.7)	-6.5 (56.5)	+0.0 (76.0)	-37.0* (40.9)
		P5 · "Which was judged more novel?"	-0.6 (73.4)	-3.9 (75.3)	-5.2 (77.3)	-2.6 (81.2)	-0.6 (83.1)	-2.6 (80.5)		-6.5 (33.1)	-2.6 (60.4)	-7.8* (78.6)	-12.3* (50.6)	-16.9* (59.1)	-18.2* (59.7)
		+ Related work (retrieval)	-1.9 (72.1)	+1.3 (80.5)	+1.3 (83.8)	-1.9 (81.8)	+0.0 (83.8)	+1.3 (84.4)		+12.3* (51.9)	+15.6* (78.6)	+0.0 (86.4)	+0.0 (63.0)	-1.3 (74.7)	-5.2 (72.7)
		Reasoning effort = low	—	+0.6 (79.9)	-2.6 (79.9)	+0.6 (84.4)	+0.0 (83.8)	-3.9 (79.2)		—	+5.8* (68.8)	-1.3 (85.1)	-3.9 (59.1)	+0.0 (76.0)	-3.2 (74.7)
		No verdict aggregation	-11.0* (63.0)	-3.9* (75.3)	-0.6 (81.8)	-3.2 (80.5)	-3.2 (80.5)	-2.6 (80.5)		-9.7* (29.9)	-8.4* (54.5)	-2.6* (83.8)	-5.2* (57.8)	-7.1* (68.8)	-3.9* (74.0)
Idea as research plan	-6.5* (67.5)	-0.7 (78.6)	-1.9 (80.5)	+1.3 (85.1)	+2.6 (86.4)	+0.0 (83.1)		-11.7* (27.9)	-4.5 (58.4)	-6.5* (79.9)	-0.7 (62.3)	-2.6 (73.4)	+1.9 (79.9)		
		sonnet-4-5	opus-4-5	opus-4-6	gpt-5.1	gpt-5.2	gpt-5.4	sonnet-4-5	opus-4-5	opus-4-6	gpt-5.1	gpt-5.2	gpt-5.4		
* p < 0.05 (two-sided paired bootstrap)															
Judge model															

* p < 0.05 (two-sided paired bootstrap)

(b) Pairwise judging, strict-accuracy (ties count as failures).

Figure 4: Cells report the change from the reference configuration, with the absolute value in parentheses. claude-sonnet-4-5 exposes no reasoning_effort parameter, hence the blank entry. Judges are **highly unstable** under even *minor* configuration changes, with large variance between models.

Appendix C.2.1).

Another disturbing pattern we observe is judge tie rates: how often a pairwise judge gives an uncertain novelty verdict (see Section 4.1). Depending on the configuration, even relatively strong judges like claude-opus-4-6 can tie on 35% of the pairs, and weaker ones like claude-sonnet-4-5 on *more than half*. Correspondingly, ignoring ties or counting them as half-correct can substantially change how different evaluation systems are ranked (see soft-accuracy results in Figure 8,

Appendix C.2). Judges tie more often on Human+Generated than on Human-Only, roughly twice as often for claude-opus-4-5 and claude-sonnet-4-5. This indicates higher uncertainty on this type of data. Turning off verdict aggregation contributes further uncertain predictions, and more than doubles tie rates in certain cases (Table 6, Appendix C.2). This, together with the tendency to tie more on generated data, explains the performance drop this change introduces on Human+Generated.

Finally, we observe that, in line with prior work

(Raina et al., 2024; Liusie et al., 2024), pairwise judging is generally more stable than pointwise judging (especially with the Human-Only data). Appendix C.2 presents results for additional performance metrics, including results without verdict aggregation and our tie-rate analysis.

5.1 Retrieval’s Limitations

Preliminary qualitative analysis. Surprisingly, retrieval contributes limited performance gains, if at all (Figure 4). To understand this further, we conduct a small-scale qualitative analysis of 10 cases where retrieval flips a correct pointwise verdict to an incorrect one (Appendix C.3). We only include ideas where the reference judge (no retrieval) is correct in all three aggregated calls and the retrieval judge is incorrect in all three, so the flips are not due to noise. We find that retrieved papers can lead the judge to dismiss contributions it previously considered novel, contrary to the human reviewers and even when nothing in the retrieved work contradicts their novelty. In one case, the judge without retrieval deemed an idea combining two concepts novel, in agreement with the reviewers. Once given papers that present each concept separately, it dismissed the same idea as “a direct combination of existing ingredients...”. Retrieval can also mislead in the opposite direction, overriding the judge’s own knowledge. Without retrieval, the judge correctly described an idea from D_- as a “broad composition of known components”, but with retrieval it concluded that the idea “...does appear novel in its overall problem formulation and synthesis *relative to the provided related work*...”. This is especially relevant, since many novelty evaluation pipelines explicitly instruct models to assess ideas against retrieved related work (Shahid et al., 2025; Moussa et al., 2026).

Stronger retrieval. A possible explanation is that our retrieval pipeline is simply too weak: with a stronger retriever, or more retrieval results, judges would ground their verdicts in the right prior work and the gains would materialize. We test this by replacing the pipeline with a substantially stronger retrieval-augmented judge and asking whether the extra retrieval quality converts into higher performance.

We instantiate a *strong agentic RAG judge* on top of gpt-5.6-sol, a newer and stronger backbone than any judge considered so far, run at reasoning_effort=xhigh and equipped with a

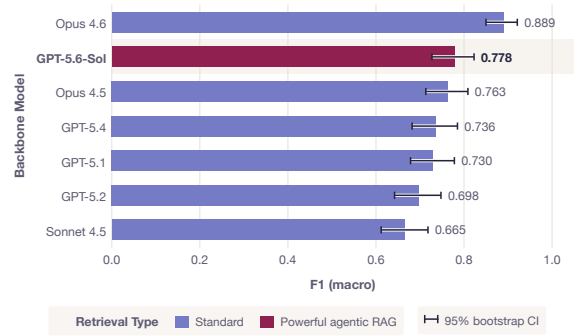


Figure 5: **Stronger retrieval does not help.** Pointwise macro-F1 on Human+Generated, for a strong agentic RAG judge (gpt-5.6-sol with live, iterative web search) against judges using the standard retrieval setup (§4), which see five pre-retrieved abstracts. All judges run once per idea, without verdict aggregation. Despite a newer backbone, higher reasoning effort, and unlimited search, the agentic judge shows no significant gains.

web-search tool. Given an idea, the judge iteratively searches the literature as it sees fit, with no cap on how many searches it issues or how many results it inspects before committing to a verdict. This differs from the pipeline described in §4, where judges are handed a fixed set of five pre-retrieved abstracts. We do limit the strong judge to prevent contamination: search is restricted to arxiv.org, and the prompt instructs the judge to consider only papers published before the retrieval cutoff (2025-03-01) and to disregard any hit that is the idea’s own preprint, mirroring a reviewer who stumbles upon the paper under review. Because this judge is costly, we call it once per idea, without verdict aggregation. Appendix B.2 gives the full implementation details.

Figure 5 compares the strong agentic RAG judge to judges using the standard retrieval pipeline (§4) on the Human+Generated set, with every judge called once per idea, without verdict aggregation, so that the two are on equal footing. The agentic judge reaches 0.78 macro-F1, 11 points behind claude-opus-4-6 with the simple five-abstract pipeline, and within noise of claude-opus-4-5 and gpt-5.4, despite costing 10× more per decision. A substantially stronger retrieval agent, with a higher reasoning budget and a newer, more powerful backbone, therefore does not improve novelty judgments, suggesting that the modest gains from retrieval in Figure 4 are not an artifact of our particular pipeline.

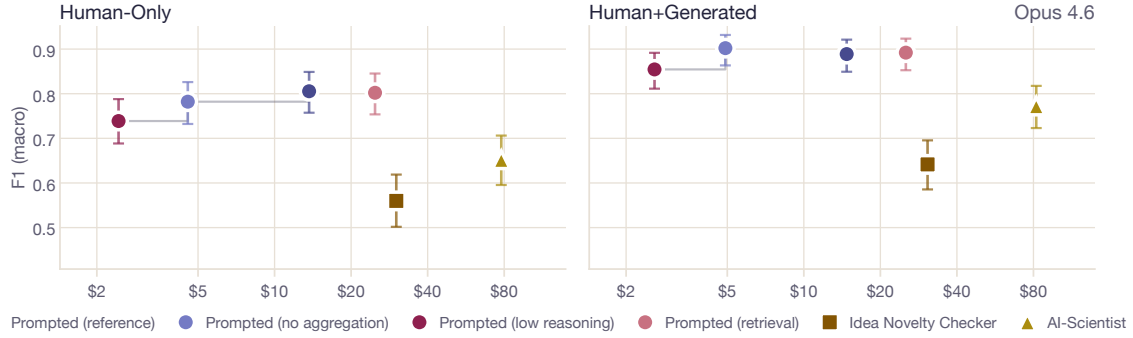


Figure 6: **Cost vs. pointwise macro-F1** for claude-opus-4-6, comparing prompted-judge configurations against the dedicated novelty judges, in the Human-Only (*left*) and Human+Generated (*right*) settings (error bars: 95% CIs from a paired bootstrap with 10,000 resamples). The Pareto front (line) contains *only* prompted judges: the *cheapest* prompted configuration outscores both dedicated judges in both settings, at over $30\times$ lower cost.

5.2 Dedicated Novelty Judges

All judges so far have been prompted judges: a vanilla LLM, a prompt, and at most a fixed set of retrieved abstracts. Prior work proposes dedicated novelty judges that spend considerably more compute and machinery per decision. In this section, we explore whether the additional compute pays off. As specified in Section 4.3, we examine Shahid et al. (2025) and Yamada et al. (2025).

We run both with claude-opus-4-6 and gpt-5.4 as backbones, under their original settings (one call per decision, without verdict aggregation). Since both are pointwise, we compare them to the prompted judges under the pointwise protocol only, and measure cost in USD over the full evaluation set. Implementation details are in Appendix B, and per-evaluator numbers in Appendix B.3.

Figure 6 plots the cost-quality tradeoff for claude-opus-4-6 (see Figure 7 for gpt-5.4). The extra machinery does not pay off: in every setting and under both backbones, our *cheapest* prompted configuration outscores both dedicated judges by 8 to 29 macro-F1 points, while costing up to $32\times$ less. The Pareto front consists entirely of simple prompted judges.

Conclusions

We ran a controlled study of automated novelty evaluation, sweeping the design choices a practitioner makes when building an LLM novelty judge. To enable this, we devised an automatic pipeline that mines novelty labels from free-text peer reviews and pairs human-authored with LLM-generated ideas, evaluating judges across both distributions.

Judges turn out to be highly sensitive: small edits to the judge prompt swing pairwise accuracy by more than 50 points, the same edit helps one backbone and sinks another below chance, and the fixes practitioners reach for first, retrieval and higher reasoning effort, buy little. Paying more does not help either: the dedicated novelty judges we tested are outscored by our cheapest prompted configuration, which operates at a fraction of their cost.

Pairwise judges in particular prove least stable in precisely the scenarios they are built to address. Changes that barely move Human-Only performance shift Human+Generated by tens of points, and judges show increased uncertainty once generated data is evaluated. Validating a judge on human-authored papers, or against a single ideation system, says little about how it will behave on the generated ideas it is actually asked to score. Until evaluation protocols account for these sensitivities, automated novelty assessments should be interpreted with significant caution.

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A Additional Benchmark Details

A.1 Human Data Collection

We begin human-authored idea collection by fetching all ICLR 2026 submissions and their corresponding reviews. We then apply claude-opus-4-6 to extract novelty signals from the raw weaknesses/strengths review sections. Prompt 1 presents the prompt used for novelty signal extraction, and Table 3 shows extraction examples. Next, we apply filters and divide the submissions into two groups as follows:

- **High-novelty papers:** *Accepted* submissions that have received an average rating within the *top* 10% of their ICLR area (and that is ≥ 6), and an average contribution score ≥ 3 . Additionally, these submissions must have *no negative novelty signals* and must receive *positive* novelty signals from the *majority* of reviewers.
- **Low-novelty papers:** *Rejected* submissions that have received an average rating within the *bottom* 10% of their ICLR area (and that is ≤ 3.5), and an average contribution score ≤ 2 . Additionally, these submissions must have *no positive novelty signals* and must receive *negative* novelty signals from the *majority* of reviewers.

We additionally exclude submissions whose primary ICLR area is *other topics in machine learning* (i.e., none of the listed areas). This category aggregates a heterogeneous mix of unrelated subfields, which makes novelty comparisons between its submissions noisy.

Finally, we apply claude-opus-4-6 to the remaining ideas to remove evaluation information related to actual results or findings specified in the paper (unless this data is key to the central contribution). Prompt 2 presents the relevant prompt for this step.

A.2 Benchmark Data Setups Construction

Human-Only. Ideas collected through the process described in Appendix A.1 are used in the pointwise setting as is, or paired according to their primary ICLR area (e.g., “*generative models*”) in the pairwise setting. Since we have more high-novelty human data than low-novelty human data, a low-novelty idea might be paired with a high-novelty idea more than once.

Human+Generated. We pair each high-novelty, human-authored idea with an LLM-generated counterpart designed to simulate standard research generation. To produce this baseline, we seed claude-sonnet-4-5 with the human idea’s ICLR area (e.g., “*generative models*”). The model is instructed to reason about potential research directions before proposing a final idea, using the prompt detailed in Prompt 3. We use identical generation parameters (max_tokens=8192, default settings). As before, these pairs are decoupled to create the pointwise data.

A.3 Expert Validation

We ran a small blind study to check that the weak-labeling assumption behind Human+Generated (Section 3.1) is not obviously violated. A computer science professor with broad expertise across AI-related domains evaluated 30 idea pairs from the Human+Generated pool. Of these, 28 were pairs that our top-performing pairwise judge (claude-opus-4-6) misclassified, and the remaining two were control pairs it classified correctly in all six verdicts. The expert did not know which pairs were controls. For each pair, the expert was asked to identify the more novel idea, provide written justifications, and cite relevant prior literature when possible. The expert’s judgment aligned with our ground-truth labels on all 30 pairs, including all 28 judge-error pairs (approximately five hours of annotation in total). Table 4 gives representative examples of these annotations.

B Additional Judge Details

B.1 Prompted Judges

All prompted judge models use max_tokens=8192. Additionally, all models (except claude-sonnet-4-5, which does not support the option) are configured with reasoning_effort="high", using default values for all other parameters. For the pointwise evaluation, we call the judge three times and take the majority vote. For the pairwise evaluation, we mitigate position bias by adopting the method from Wang et al. (2023) and calling the judge three times per presentation order (A vs. B and B vs. A). When a judge selects an idea as the winner, it is assigned a score of 1, while the losing idea receives a 0. The final winner is determined by the higher average score, and if both ideas receive the same average, the match is declared a

Novelty Signals Extraction Prompt

Role

You are an expert Research Idea Auditor specializing in peer-review analysis.

Task

Identify and extract verbatim snippets from the provided review sections that evaluate the novelty, originality, or uniqueness of the research. Classify each snippet as **positive** or **negative** based on its meaning regardless of which section (Strengths or Weaknesses) it appears in.

Criteria for Classification

- * **Positive Novelty/Originality:** Only extract explicit originality claims about the contribution itself (e.g., "new," "first," "pioneering," "unique," "original," "novel"). The text must directly state novelty relative to prior work or standard methods.
- * **Negative Novelty (Lack of Originality):** Only extract when the text's primary claim explicitly judges the contribution itself as non-novel (e.g., "incremental," "derivative," "already known," "standard," "marginal improvement," "not novel," "not very novel"). A sentence found in the Strengths section that negatively judges novelty belongs in `negative\_novelty\_snippets`.

Exclusions (Do NOT Extract)

- * **Generic Praise/Opinion:** Any sentence that praises quality, elegance, usefulness, efficiency, or clarity without an explicit novelty reference.
- * **Capability/Property Claims:** Descriptions of what the method does, its scope, or its properties without explicitly claiming this was not done before. This includes sentences using verbs like "expands," "extends," "applies," "explores," "studies," "evaluates," or "investigates" that describe the paper's actions rather than asserting priority (e.g., "generalizes existing methods," "approximates learning dynamics," "unifies X and Y perspectives," "expands scaling law studies to generative evaluations"). A sentence must assert **first-time** or **prior-absence** to qualify describing what was done is not sufficient.
- * **Literature Positioning Process:** Text about missing citations, omitted prior work, or how claims are communicated unless they directly state the contribution is already known.
- * **Experiment Critique:** Sentences about marginal gains, weak improvements, or missing comparisons unless they explicitly conclude the contribution itself is non-novel.
- * **Reviewer Uncertainty:** Sentences where the reviewer expresses inability to assess novelty due to missing context these reflect an incomplete review, not a verdict that the contribution is non-novel. Key signal: hedging language such as "difficult to say," "hard to assess," "cannot determine," "unclear if," "it is hard to tell," or conditional framing like "without X, I cannot judge Y." Example: "Without a related work section, it is difficult to say this paper is novel" do NOT extract this; the reviewer is withholding judgment, not rendering one.

Litmus Test

Before extracting a positive snippet, ask: **"Does this sentence explicitly say the contribution did not exist before, or was not done before?"** If the answer is no, do not extract it. Describing **what** the paper does (even in a new domain or setting) is not the same as claiming it has never been done do not extract the former.

Before extracting a negative snippet, ask: **"Does this sentence explicitly judge the contribution itself as non-novel, rather than merely noting weak results, missing experiments, poor presentation, or the reviewer's own inability to assess novelty?"** If the answer is no, do not extract it.

Constraints

- * **Verbatim Only:** Do not paraphrase. Extract the exact text from the source.
- * **Classify by meaning, not by section:** A snippet from Strengths that negatively judges novelty goes into `negative\_novelty\_snippets`, and vice versa.
- * **Empty State:** If no text snippets meet the criteria, return an empty list `[]`.

Input Data

Strengths Section:
{ { strengths_review } }

Weaknesses Section:
{ { weaknesses_review } }

Output Format

Return a JSON object with the following structure:

```
{
  "positive_novelty_snippets": ["list of exact strings"],
  "negative_novelty_snippets": ["list of exact strings"],
}
```

Prompt 1: Novelty signals extraction prompt.

Remove Evaluation Information Prompt

****Role**:**

You are an expert academic editor. Your task is to produce a modified version of a research paper abstract that omits all evaluation-related information.

****Input Abstract**:**

{{ abstract }}

****What to Remove**:**

Remove information that validates or benchmarks the proposed method against others, including:

- Specific datasets or benchmarks used *only* to validate the method (e.g., "we evaluate on SQuAD and GLUE").
- Quantitative results or performance metrics (e.g., accuracy, F1, BLEU scores).
- Comparisons to baselines or prior methods.
- URLs or references to code repositories or supplementary materials.

****What to Keep**:**

- Context, motivation, and problem statement.
- The description of the proposed method or system.
- Datasets, benchmarks, or analyses that *are* the main contribution (e.g., a newly introduced benchmark, a dataset curated as part of the work, or an empirical analysis that is the paper's primary goal).

****Key Distinction**:**

Ask yourself is the dataset/evaluation *serving* the contribution, or *is it* the contribution?

- "We evaluate our model on ImageNet" remove (validation of the method).
- "We introduce a dataset of 10K annotated examples" keep (the dataset is the contribution).
- "We present the first analysis of X using Y dataset" keep (the analysis is the contribution).

****Critical Rules**:**

- Change as little as possible preserve the original wording everywhere else.
- The revised abstract must remain fluent and self-contained; bridge any gaps left by removed sentences naturally.
- The result must still read like an abstract from a top-tier venue (e.g., ACL, ICLR, NeurIPS).
- Return only the revised abstract text, with no additional commentary or formatting.

Prompt 2: Remove evaluation information prompt.

ICLR 2026 submission	Extracted reviewer novelty signal examples
Positive novelty signals	
<i>Image Quality Assessment for Embodied AI</i>	“The paper identifies and clearly defines a completely new, critical, and timely research problem ”
<i>Quotient-Space Diffusion Models</i>	“It is the first work I have seen that attempts to constrain the diffusion process within the quotient space ”
<i>Critical Confabulation: Can LLMs Hallucinate for Social Good?</i>	“this paper is so unlike most of the ICLR paper I’ve seen in recent years , especially with its use of historical archival data... ”
Negative novelty signals	
<i>DFCA: Decentralized Federated Clustering Algorithm</i>	“While the theoretical contribution is not highly novel , the paper provides a useful convergence discussion...”
<i>NoLoRA: Nonlinear Low-Rank Adaptation for Parameter-Efficient Fine-Tuning</i>	“The proposal is extremely close to NEAT (Zhong et al., 2025). Both methods replace LoRA’s linear update with a nonlinear mapping...”
<i>Consistent 3D Object Detection with Active LLM Reasoning</i>	“The contribution is primarily procedural integration of existing components (YOLOE + CLIP + GPT-4V + SAM2), with no novel learning formulation or analytical insight... ”

Table 3: **Novelty signals extracted from ICLR 2026 reviews.**

tie. Prompts 4 and 5 give the reference pairwise and pointwise prompts respectively.

B.2 Strong Retrieval Judge

The strong retrieval judge introduced in §5.1 is built on gpt-5.6-sol. We set reasoning_effort to xhigh and enable the provider-hosted web_search tool, restricted to arxiv.org. We impose no cap on the number of tokens used, so the judge is free to issue as many searches and inspect as many results as it deems necessary before committing to a verdict. Given its higher price, we call the judge once per idea, without verdict aggregation. The prompt for this judge is given in Prompt 6. It is identical to the reference prompt (Prompt 5), except for additional search-tool-related instructions. The added guidance constrains the search to papers published before the retrieval cutoff (2025-03-01) and instructs the judge to disregard any hit that is the idea’s own preprint, mirroring a human reviewer who happens to stumble upon the paper under review.

B.3 Dedicated Novelty Judges

We follow the implementations released by Shahid et al. (2025)⁴ for both judges. Each is evaluated with claude-opus-4-6 and gpt-5.4 as its backbone, at reasoning_effort “high” and a single call per decision, without verdict aggregation. Unlike our prompted judges, both perform their own

Semantic Scholar retrieval rather than receiving a fixed candidate set. We restrict those searches to papers published before the retrieval cutoff (2025-03-01) and discard any hit whose title matches the paper the idea was drawn from.

We use the following configurations for the judges:

- **Idea Novelty Checker.** We keep the upstream defaults: keyword, title, and snippet search for paper collection, Specter2 embedding filtering, a RankGPT rerank in its priority variant, and a verdict conditioned on the top 10 ranked papers, together with the relaxed in-context example set. The rerank is run with gpt-5.4-mini under both backbones.
- **AI-Scientist.** We allow the agent up to 10 search rounds, the upstream default, feeding each round’s results into the next round’s prompt.

Additional results. Table 5 reports the full per-judge numbers behind Figure 6, and Figure 7 gives the corresponding cost-quality view for gpt-5.4. Under both backbones and in both settings, the dedicated judges are the most expensive configurations we run and the least accurate, i.e., they are not cost-effective.

⁴https://github.com/simra-shahid/idea_novelty_checker

Idea Generation Prompt

You are a scientific idea generator. Given a research area, generate a novel idea.

****Instructions**:**

- Reason about the given area and possible research directions.
- Write your final idea as a research paper abstract.

****Abstract Writing Guidelines**:**

- Write a fluent, unstructured abstract in the style of a top-tier venue (e.g., NeurIPS, ICLR, ACL).
- No Evaluation: Do not include evaluation results, benchmark comparisons, quantitative metrics, or dataset sizes.
- Length: The abstract must not exceed 200 words.

Output Format Example:

- reasoning: "..."
- abstract: "Foundation models for Earth Observation produce rich general-purpose representations, yet they lack the domain-specific ecological knowledge needed for biodiversity and environmental modeling tasks. Adapting such models to incorporate ecological expertise without full retraining remains an open challenge, particularly when catastrophic forgetting and computational cost must be minimized. BotaCLIP is a lightweight multimodal contrastive framework that aligns high-resolution aerial imagery with botanical relev  data on top of a pre-trained Earth Observation foundation model. Contrastive learning injects ecological structure into the representation space, while a regularization strategy preserves the original model's general knowledge. The resulting embeddings serve as transferable inputs for downstream ecological predictors spanning plant presence, species occurrence, and soil composition tasks."

****Research Area**:**

{{ area }}

Prompt 3: Idea generation prompt.

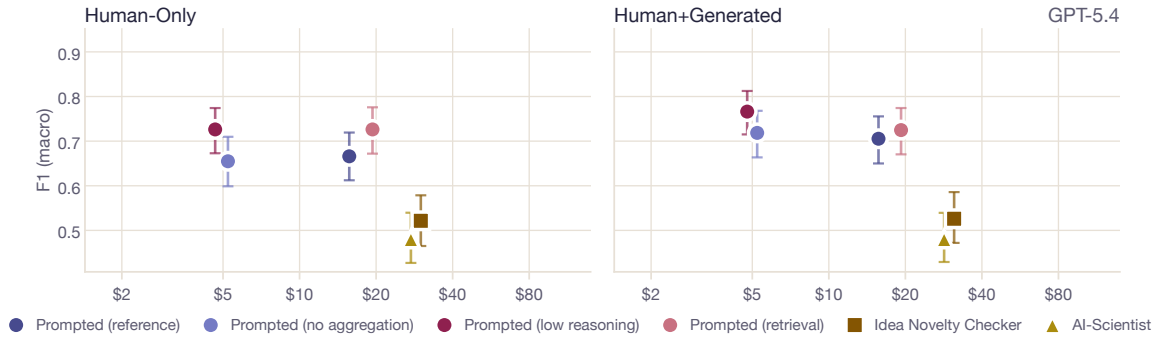


Figure 7: **Cost vs. pointwise macro-F1** for gpt-5.4, the counterpart of Figure 6, comparing prompted-judge configurations against the dedicated novelty judges, in the Human-Only (left) and Human+Generated (right) settings. The same pattern holds: every prompted configuration outscores both dedicated judges, at roughly 6× lower cost (error bars: 95% CIs from a paired bootstrap with 10,000 resamples).

C Controlled Evaluation Study Details

C.1 Implementation Details

Judge prompt. Five variants replace the reference prompts (Prompts 4 and 5): P1 and P2 in both the pairwise and the pointwise setting, P3–P5

in the pairwise setting only. *P1: Criteria without guardrails* strips the novelty criterion of its guardrails against trivial novelty, deleting the three paragraphs shown in Prompt 7. *P2: Criteria removed* goes further and drops the criterion alto-

Reference Pairwise Judge Prompt

You are a senior reviewer for an AI scientific conference such as ICLR, ACL, ICML. Your task is to review the novelty of the following two ideas and to decide which one is more novel. Focus only on the novelty and not on other aspects such as feasibility, clarity, etc.

Idea-0

IDEA 0

Idea-1

IDEA 1

Compare these ideas based on the following criteria:

Novelty: The substantive originality of the idea, characterized by:

- The introduction of new frameworks, concepts, evaluations, resources or approaches absent from existing knowledge.
- The non-trivial synthesis of existing frameworks, concepts, evaluations, resources or approaches.
- The extension of established frameworks, concepts, evaluations, resources or approaches into previously unexplored problem domains.

In general, composite methods / multi-stage pipelines built from several existing motifs in haphazard manner, stringing together all kinds of common methods, are NOT to be considered novel, except if the composition is highly different than any existing work, involves substantial creativity, and the combination is highly non-trivial.

Do NOT rely on assumptions of all kinds of unknown hypotheticals. For example, "this could be novel IF formulated as..." is NOT acceptable as a reason to judge an idea as novel. Only use what is explicitly stated.

Do NOT assume a new model/training method/technical algorithmic/mathematical contribution is more novel than a new analysis, new evaluation, new task, or new resource. All types of contributions could either be novel, or not, depending on context and the state of scientific knowledge.

****Tip****: Think about the main contributions of each idea. Which makes more novel contributions? Which makes relatively more incremental contributions?

****Comparative Logic****

- * Assign ****0**** if the novelty of Idea-0 is superior.
- * Assign ****1**** if the novelty of Idea-1 is superior.

****Required Response Format****

<thinking>

[Your reasoning here]

</thinking>

```json

```
{
 "novelty": <0 or 1>
}
```

```

Prompt 4: **Reference pairwise judge prompt**, shown exactly as sent to the judge apart from the two **IDEA** placeholders, which hold the ideas under comparison.

Reference Pointwise Judge Prompt

You are a senior reviewer for an AI scientific conference such as ICLR, ACL, ICML. Your task is to review the novelty of the following idea and to decide whether it is novel. Focus only on the novelty and not on other aspects such as feasibility, clarity, etc.

Idea

IDEA

Judge this idea based on the following criteria:

Novelty: The substantive originality of the idea, characterized by:

- The introduction of new frameworks, concepts, evaluations, resources or approaches absent from existing knowledge.
- The non-trivial synthesis of existing frameworks, concepts, evaluations, resources or approaches.
- The extension of established frameworks, concepts, evaluations, resources or approaches into previously unexplored problem domains.

In general, composite methods / multi-stage pipelines built from several existing motifs in haphazard manner, stringing together all kinds of common methods, are NOT to be considered novel, except if the composition is highly different than any existing work, involves substantial creativity, and the combination is highly non-trivial.

Do NOT rely on assumptions of all kinds of unknown hypotheticals. For example, "this could be novel IF formulated as..." is NOT acceptable as a reason to judge an idea as novel. Only use what is explicitly stated.

Do NOT assume a new model/training method/technical algorithmic/mathematical contribution is more novel than a new analysis, new evaluation, new task, or new resource. All types of contributions could either be novel, or not, depending on context and the state of scientific knowledge.

**** Review Logic:****

* Assign ****0**** if the idea is not novel.

* Assign ****1**** if the idea is novel.

****Required Response Format:****

<thinking>

[Your reasoning here]

</thinking>

```json

{

  "novelty": <0 or 1>

}

```

Prompt 5: **Reference pointwise judge prompt.** The pointwise counterpart of Prompt 4.

Strong Retrieval Judge Prompt

You are a senior reviewer for an AI scientific conference such as ICLR, ACL, ICML. Your task is to review the novelty of the following idea and to decide whether it is novel. Focus only on the novelty and not on other aspects such as feasibility, clarity, etc.

Before making your judgment, use your web search tool to find ArXiv papers relevant to the given idea.

Idea

IDEA

Judge this idea based on the following criteria:

Novelty: The substantive originality of the idea, characterized by:

- The introduction of new frameworks, concepts, evaluations, resources or approaches absent from existing knowledge.
- The non-trivial synthesis of existing frameworks, concepts, evaluations, resources or approaches.
- The extension of established frameworks, concepts, evaluations, resources or approaches into previously unexplored problem domains.

In general, composite methods / multi-stage pipelines built from several existing motifs in haphazard manner, stringing together all kinds of common methods, are NOT to be considered novel, except if the composition is highly different than any existing work, involves substantial creativity, and the combination is highly non-trivial.

Do NOT rely on assumptions of all kinds of unknown hypotheticals. For example, "this could be novel IF formulated as..." is NOT acceptable as a reason to judge an idea as novel. Only use what is explicitly stated.

Do NOT assume a new model/training method/technical algorithmic/mathematical contribution is more novel than a new analysis, new evaluation, new task, or new resource. All types of contributions could either be novel, or not, depending on context and the state of scientific knowledge.

**** Review Logic:****

* Assign ****0**** if the idea is not novel.

* Assign ****1**** if the idea is novel.

****Search Guidelines****

Use your web search tool to find relevant arXiv papers to inform your novelty judgment. You are constrained to search papers before a given date: **CUTOFF DATE**. Do NOT search or use any papers beyond this date. I want you to be very careful about not seeing ANY papers after **CUTOFF DATE**. Apply guardrails to your queries as much as possible to minimize the chance of seeing any papers after the **CUTOFF DATE**.

For keyword searches, always include an explicit date filter such as ``before:CUTOFF DATE``. For direct arXiv lookups, only reference papers whose ID prefix encodes a date before the cutoff (e.g. ``arxiv.org/abs/2503.XXXXX`` for March 2025).

If you come across a paper that appears to be the idea itself (i.e. essentially an exact match describing the same method, framing, and contributions --- as if the authors uploaded their own paper for this idea to arXiv), completely ignore it and do not treat it as prior work. Like a human reviewer searching for related work who happens to stumble upon the very paper under review, you must disregard it when judging novelty.

List all the queries you used.

****Required Response Format:****

`<thinking>`

[Your reasoning here]

`</thinking>`

````json`

```
{
 {"novelty": <0 or 1>}
}
```

`````

Prompt 6: Strong retrieval judge prompt. The judge searches the literature itself instead of being handed a fixed candidate list.

Deleted from Both Reference Prompts

In general, composite methods / multi-stage pipelines built from several existing motifs in haphazard manner, stringing together all kinds of common methods, are NOT to be considered novel, except if the composition is highly different than any existing work, involves substantial creativity, and the combination is highly non-trivial.

Do NOT rely on assumptions of all kinds of unknown hypotheticals. For example, "this could be novel IF formulated as..." is NOT acceptable as a reason to judge an idea as novel. Only use what is explicitly stated.

Do NOT assume a new model/training method/technical algorithmic/mathematical contribution is more novel than a new analysis, new evaluation, new task, or new resource. All types of contributions could either be novel, or not, depending on context and the state of scientific knowledge.

Prompt 7: **P1: Criteria without guardrails, diff against Prompts 4 and 5.** P1 deletes the three paragraphs shown, which constrain what counts as novel.

gether (Prompts 8 and 9). P3–P5 are not edits of the reference prompt but a separate lineage, adapted from Si et al. (2024), whose original prompt asks which of two abstracts was accepted at a conference. P3: "Which was judged novel?" presumes that a prior review found exactly one of the two ideas novel and asks the judge to predict which (Prompt 10), P4: "Which is novel?" drops that presumption (Prompt 11), and P5: "Which was judged more novel?" restores it in a comparative form (Prompt 12).

Idea format. We use claude-opus-4-6 to extract a two-field structured plan from the free-form idea, following the prompt detailed in Prompt 13. We then treat the resulting plans as the ideas, and give them to the judges as described in Appendix B.1.

Retrieval. We conduct retrieval for each benchmark idea by first extracting up to three main contributions using claude-opus-4-6 with the prompt specified in Prompt 14. Following this extraction, we instruct the model to simulate the thought pro-

cess of a novelty reviewer and generate five targeted queries designed to probe the novelty of these specific contributions, utilizing the prompt detailed in Prompt 15. We then leverage the queries to conduct a Paper-Finder⁵ search, and retrieve initial candidates. Next, we filter out the ones published after the retrieval cutoff (2025-03-01). For each query, we keep the remaining candidate with the highest relevance score, yielding 5 abstracts as the final retrieved related work. Prompt 16 shows how those abstracts enter the judge prompt, and the accompanying rewording of the novelty criterion, which asks about originality relative to the supplied prior work. We show the pairwise criterion; the pointwise criterion is reworded similarly.

Significance testing. We test the significance of every change reported relative to the reference configuration (Section 4.1) with a two-sided paired non-parametric bootstrap of 100,000 resamples. For a given judge and metric, we resample the evaluation instances with replacement, apply the same resampled indices to the predictions of the reference and of the modified configuration, and recompute the difference between the two. The bootstrap is corpus-level: on each resample we recompute the metric (macro-F1, soft-accuracy, or strict-accuracy) over the full resampled set. From the resulting distribution of differences we build a bias-corrected and accelerated (BCa) 95% confidence interval, and mark a difference with * when this interval excludes zero (i.e., $p < 0.05$).

C.2 Additional Results

Soft accuracy results. Figure 8 repeats the pairwise experiment in Figure 4b under *soft* accuracy, where ties count as half-correct. Judges obtain higher results under the soft metric, but the ablation effects follow the same patterns, with high variance and with the Human-Only data being less sensitive to perturbations.

Per-class pointwise results. Figure 9 reports ablation results using F1 on the *novel* class (Figure 9a) and F1 on the *not-novel* class (Figure 9b).

Tie rates. Figure 10 reports the tie rate of each pairwise judge across the configurations presented in Figure 4b (roughly 20 data points per judge). Table 6 shows the exact tie rate per configuration and model. We observe that judges tie more on average once D_- is LLM-generated, with

⁵<https://github.com/allenai/asta-paper-finder>

P2 Pairwise Judge Prompt

You are a senior reviewer for an AI scientific conference such as ICLR, ACL, ICML. Your task is to review the novelty of the following two ideas and to decide which one is more novel. Focus only on the novelty and not on other aspects such as feasibility, clarity, etc.

Idea-0
IDEA 0

Idea-1
IDEA 1

****Comparative Logic:****
* Assign ****0**** if the novelty of Idea-0 is superior.
* Assign ****1**** if the novelty of Idea-1 is superior.

****Required Response Format:****
<thinking>
[Your reasoning here]
</thinking>

```json  
{"novelty": <0 or 1>}  
```

Prompt 8: **P2: Criteria removed, pairwise.**

P2 Pointwise Judge Prompt

You are a senior reviewer for an AI scientific conference such as ICLR, ACL, ICML. Your task is to review the novelty of the following idea and to decide whether it is novel. Focus only on the novelty and not on other aspects such as feasibility, clarity, etc.

Idea
IDEA

****Review Logic:****
* Assign ****0**** if the idea is not novel.
* Assign ****1**** if the idea is novel.

****Required Response Format:****
<thinking>
[Your reasoning here]
</thinking>

```json  
{"novelty": <0 or 1>}  
```

Prompt 9: **P2: Criteria removed, pointwise.**

P3 Judge Prompt (Si et al., 2024, Modified for novelty evaluation)

You are a reviewer specialized in Artificial intelligence and Machine Learning research. You are given two ideas. One of them is judged as novel by reviewers for a top AI conference (like ICLR or ACL) reviewers and the other one is not. Your task is to identify the one that is judged as novel.

The two ideas are:

```
paper 0: {{ idea0.text }}  
paper 1: {{ idea1.text }}
```

Now decide which one is judged as novel. Return a json string with the selected idea index {'novelty': <index>} and nothing else.

Prompt 10: **P3: “Which was judged novel?”**, adapted from Si et al. (2024), whose original prompt asks which of two abstracts was accepted at a conference.

P4 Judge Prompt

You are a reviewer specialized in Artificial intelligence and Machine Learning research. You are given two ideas. Your task is to identify the one that is novel.

The two ideas are:

```
paper 0: {{ idea0.text }}  
paper 1: {{ idea1.text }}
```

Now decide which one is novel. Return a json string with the selected idea index {'novelty': <index>} and nothing else.

Prompt 11: **P4: “Which is novel?”**, which drops P3’s reviewer and venue framing.

P5 Judge Prompt

You are a reviewer specialized in Artificial intelligence and Machine Learning research. You are given two ideas. One of them is judged as more novel than the other by reviewers for a top AI conference (like ICLR or ACL). Your task is to identify the idea that is judged as more novel.

The two ideas are:

```
paper 0: {{ idea0.text }}  
paper 1: {{ idea1.text }}
```

Now decide which one is judged as more novel. Return a json string with the selected idea index {'novelty': <index>} and nothing else.

Prompt 12: **P5: “Which was judged more novel?”**, which replaces P3’s binary novel/not-novel contrast with a graded one.

Research Plan Extraction Prompt

****Role**:**

You are an expert technical data extractor. Your task is to extract a concise , structured research plan from a research paper abstract.

****Input Text**:**

{{ abstract }}

****Field Definitions**:**

- purpose: The goal or objective of the research. May include (but is not limited to):
 - The problem the research aims to solve
 - The research question it addresses
 - The gap or limitation it targets
 - The capability it seeks to enable or improve.
- mechanism: The approach used to achieve the purpose. May include (but is not limited to):
 - The algorithm, model, or procedure proposed
 - The dataset or benchmark constructed
 - The theoretical framework or formal analysis applied
 - The experimental methodology or evaluation protocol introduced.

****Constraints**:**

- Do not include specific numbers, counts, percentages, or dataset sizes (e.g . avoid "28K records" or "top-5% accuracy").
- Omit language such as "this paper", "the work", or "we propose", and focus on the technical content of the research plan.
- Write in plain, precise language a human should be able to read the two fields together and fully understand the research plan without consulting the abstract.
- Each field should form a complete, standalone section that contributes to a coherent narrative when read in sequence.
- The total word count across all fields must not exceed 200 words.

****Important**:** If a field is not explicitly stated in the abstract, return an empty string ("") for that field. Do not infer, deduce, or add any information beyond what is directly written in the abstract.

****Output Schema**:**

Return only a valid JSON object with the root key "idea". Do not include markdown formatting (e.g. ```json) or conversational text.

```
{
  "idea": {
    "purpose": "String",
    "mechanism": "String",
  }
}
```

Prompt 13: Research plan extraction prompt.

Retrieval Contribution Prompt

```
===== SYSTEM =====

### ROLE
Your goal is to extract structured, high-fidelity scientific contributions
from research ideas to facilitate literature reviews and automated search
query generation.

### TASK
Identify the core dimensions of contribution and extract specific, context-
rich statements for each.

### INSTRUCTIONS
- **Identify Contribution Dimensions**: Extract high-level categories that
are meaningful and comparable across research fields. While you should
adapt these to the specific idea, prioritize these standard pillars:
  - **Methodology**: New algorithms, models, procedures, or technical
  protocols.
  - **Application**: Novel deployment of existing methods to a new problem
  or domain.
  - **Theoretical**: New proofs, mathematical derivations, or formal
  theorems.
  - **Data/Resource**: Novel datasets, original surveys, or unique
  measurement corpora.
  - **Evaluation/Benchmark**: New experimental designs, metrics, or
  comparative frameworks.
  - **Systems/Tools**: New software, hardware architectures, or research
  infrastructure.
  - **Conceptual**: New taxonomies, ontologies, or foundational mental
  models.
  - **Findings/Analysis**: New empirical findings, derived conclusions, or
  novel analyses.
  - **Problem Formulation**: New research questions, problem definitions,
  or framing.
- **Generate Contribution Statements**: For each dimension, write one or more
specific contribution statements. These statements should be precise,
self-contained, and informative make sure they include enough context as
they will be used as the basis to generate search queries later on.

### OUTPUT
Please output a parseable JSON block as follows:

'''json
{
  "<dimension_name_1>": [
    "<contribution_statement_1>",
    "<contribution_statement_2>"
  ],
  "<dimension_name_2>": [
    "<contribution_statement_3>"
  ]
}

===== USER =====

Research Idea:
{{ idea_text }}

Please Identify a maximum of {{ n_contributions }} contributions.
```

Prompt 14: Retrieval contribution prompt.

Retrieval Query Generation Prompt

```
===== SYSTEM =====

**Role**: You are a senior reviewer for a top-tier AI conference (e.g., ICLR,
NeurIPS, ACL).

**Context**: You will receive a research idea and a summary of its main
contributions. You have access to a sophisticated scientific search engine
that retrieves relevant papers given nuanced, natural language queries.
Your goal is to generate search queries that surface papers useful for
evaluating how novel the proposed research is relative to prior work.

**Task**: Generate up to {{ n_queries }} search queries.

Guidelines:
* Each query should be brief and focused.
* Avoid generating queries that are too general, overly broad, or likely to
  return unrelated results.
* Do not only generate queries that are too specific to the research idea
  you are not only looking for papers that do the exact same thing, but also
  ones that are more broadly related or share meaningful overlap with parts
  of the idea.
* Reflect an understanding of the broader research idea, but do not drift
  away from the specific research context.
* Write queries in nuanced, natural language not keyword searches. Phrase
  them as sentences or descriptions (e.g., "Papers about...", "Research on
  ...", "Methods for...") rather than lists of keywords.
* Ensure queries are meaningfully different from one another.
* Try to cover as many contributions as possible with your queries.

===== USER =====

Research Idea:
{{ idea_text }}

Contributions:
{{ contributions_summary }}

Return a JSON object with a single key 'queries' which is a list of strings.
```

Prompt 15: Retrieval query generation prompt.

Idea Blocks, With Related Work Appended

Idea-0

IDEA 0

[Start Related Work for idea0]

RETRIEVED ABSTRACTS

[End Related Work for idea0]

Idea-1

IDEA 1

[Start Related Work for idea1]

RETRIEVED ABSTRACTS

[End Related Work for idea1]

Criterion, Reworded

Novelty: The substantive originality of the idea **in relation to the provided related work**, characterized by:

...

****Tip****: Think about the main contributions of each idea. Which makes more novel contributions **over prior work**? Which makes relatively more incremental contributions **over prior work**?

Prompt 16: **Retrieval, diff against Prompts 4 and 5**. *Top*: the retrieved abstracts are appended to each idea inside a delimited block, shown here for the pairwise reference prompt; the pointwise reference prompt takes a single such block, with the delimiters reading [Start Related Work] and [End Related Work]. *Bottom*: the criterion picks up the three highlighted insertions, so that it asks about originality relative to the supplied prior work. The elided paragraphs and the rest of the prompt are unchanged.

| | | Human-only | | | | | | Human + generated | | | | | |
|--------|-------------------------------------|-----------------|----------------|-----------------|------------------|----------------|-----------------|-------------------|------------------|-------------------|------------------|------------------|------------------|
| Change | Reference | 80.2 | 81.5 | 82.8 | 87.0 | 85.4 | 85.4 | 57.1 | 72.1 | 88.0 | 66.6 | 77.3 | 80.5 |
| | P1 · Criteria without guardrails | -1.3
(78.9) | +1.0
(82.5) | -0.6
(82.1) | -1.6
(85.4) | -1.3
(84.1) | -3.2
(82.1) | -12.7*
(44.5) | -13.3*
(58.8) | -4.9*
(83.1) | -9.4*
(57.1) | -8.8*
(68.5) | -10.4*
(70.1) |
| | P2 · Criteria removed | -2.3
(77.9) | +2.6
(84.1) | +0.7
(83.4) | -1.3
(85.7) | -0.3
(85.1) | -1.0
(84.4) | -4.9*
(52.3) | -2.6
(69.5) | -1.9
(86.0) | -4.9*
(61.7) | +0.0
(77.3) | +0.6
(81.2) |
| | P3 · "Which was judged novel?" | +0.7
(80.8) | -0.3
(81.2) | +1.3
(84.1) | +0.3
(87.3) | +0.7
(86.0) | +0.0
(85.4) | +9.1*
(66.2) | +23.4*
(95.5) | +12.0*
(100.0) | +15.9*
(82.5) | +7.1*
(84.4) | +14.3*
(94.8) |
| | P4 · "Which is novel?" | -4.5
(75.6) | -1.6
(79.9) | -7.1*
(75.6) | -11.0*
(76.0) | -3.2
(82.1) | -7.1*
(78.2) | -12.7*
(44.5) | +4.9
(77.0) | -1.9
(86.0) | -6.5
(60.1) | +1.0
(78.2) | -34.7*
(45.8) |
| | P5 · "Which was judged more novel?" | -1.3
(78.9) | -0.6
(80.8) | -1.3
(81.5) | -3.9
(83.1) | -1.3
(84.1) | -4.2
(81.2) | -13.0*
(44.2) | -0.3
(71.8) | -1.3
(86.7) | -11.7*
(54.9) | -15.3*
(62.0) | -17.9*
(62.7) |
| | + Related work (retrieval) | +0.7
(80.8) | +3.3
(84.7) | +3.3
(86.0) | -1.9
(85.1) | +0.3
(85.7) | +1.9
(87.3) | +10.7*
(67.9) | +10.7*
(82.8) | +0.3
(88.3) | -1.9
(64.6) | +0.3
(77.6) | -4.2
(76.3) |
| | Reasoning effort = low | — | +1.0
(82.5) | -1.0
(81.8) | -1.0
(86.0) | +0.0
(85.4) | -3.2
(82.1) | — | -0.3
(71.8) | +0.3
(88.3) | -4.9*
(61.7) | +1.0
(78.2) | -4.2*
(76.3) |
| | No verdict aggregation | -3.6*
(76.6) | +0.0
(81.5) | -0.3
(82.5) | -0.6
(86.4) | -1.0
(84.4) | -0.6
(84.7) | -0.3
(56.8) | -1.0
(71.1) | -1.0
(87.0) | -1.3
(65.3) | -2.6*
(74.7) | -1.6
(78.9) |
| | Idea as research plan | -3.6*
(76.6) | -0.3
(81.2) | -0.3
(82.5) | +0.0
(87.0) | +2.6
(88.0) | -1.6
(83.8) | -5.2*
(51.9) | -6.2*
(65.9) | -4.9*
(83.1) | -1.6
(64.9) | -0.3
(77.0) | +1.3
(81.8) |
| | | sonnet-4-5 | opus-4-5 | opus-4-6 | gpt-5-1 | gpt-5-2 | gpt-5-4 | sonnet-4-5 | opus-4-5 | opus-4-6 | gpt-5-1 | gpt-5-2 | gpt-5-4 |

* p < 0.05 (two-sided paired bootstrap)

Judge model

Figure 8: Pairwise ablation results under *soft*-accuracy, where ties count as half-correct. Each cell reports the change relative to the reference configuration (top row), with the resulting absolute *soft*-accuracy in parentheses. Asterisks mark statistically significant changes.

| | | Human-only | | | | | | Human + generated | | | | | |
|--------|----------------------------------|------------------|------------------|-----------------|----------------|------------------|------------------|-------------------|-----------------|-----------------|----------------|------------------|------------------|
| Change | Reference | 50.2 | 68.1 | 81.5 | 76.3 | 55.8 | 59.2 | 60.7 | 75.3 | 87.9 | 74.4 | 54.1 | 62.5 |
| | P1 · Criteria without guardrails | +28.2*
(78.4) | +11.4*
(79.6) | -1.6
(79.9) | -0.6
(75.7) | +12.7*
(68.4) | +14.7*
(74.0) | +21.2*
(81.9) | +2.3
(77.6) | +1.8
(89.8) | +0.8
(75.2) | +11.5*
(65.6) | +10.8*
(73.3) |
| | P2 · Criteria removed | +18.4*
(68.6) | +12.2*
(80.3) | +2.4
(84.0) | +0.5
(76.7) | -9.0*
(46.7) | -8.8*
(50.5) | +13.7*
(74.4) | +8.4*
(83.7) | +4.6*
(92.6) | +2.5
(76.9) | -7.8*
(46.4) | -9.2*
(53.3) |
| | + Related work (retrieval) | +13.8*
(64.1) | +6.5*
(74.6) | -0.3
(81.3) | -1.5
(74.8) | +3.8
(59.6) | +9.3*
(68.5) | +0.6
(61.3) | -2.6
(72.7) | +0.3
(88.3) | +0.0
(74.5) | +8.3*
(62.4) | +4.2
(66.7) |
| | Reasoning effort = low | — | +2.9
(71.1) | -8.6*
(72.9) | -1.6
(74.6) | +0.3
(56.0) | +9.3*
(68.5) | — | +2.5
(77.8) | -4.5*
(83.5) | +3.4
(77.8) | +5.3*
(59.4) | +9.3*
(71.8) |
| | No verdict aggregation | -0.5
(49.8) | +1.7
(69.8) | -2.3*
(79.2) | -3.1
(73.1) | -6.7*
(49.1) | -1.5
(57.8) | -3.2
(57.5) | -1.4
(73.9) | +1.6
(89.5) | +2.5
(76.9) | +1.6
(55.7) | +2.1
(64.6) |
| | Idea as research plan | +3.2
(53.4) | +2.9
(71.0) | -1.0
(80.5) | +0.7
(77.0) | -10.0*
(45.7) | -5.2
(54.0) | -3.9
(56.8) | +0.9
(76.2) | +0.7
(88.7) | +1.4
(75.8) | -10.0*
(44.1) | -12.7*
(49.8) |
| | | | sonnet-4-5 | opus-4-5 | opus-4-6 | gpt-5.1 | gpt-5.2 | gpt-5.4 | sonnet-4-5 | opus-4-5 | opus-4-6 | gpt-5.1 | gpt-5.2 |
| | | Judge model | | | | | | | | | | | |

* p < 0.05 (two-sided paired bootstrap)

* p < 0.05 (two-sided paired bootstrap)

(a) F1 on the **novel** class.

| | | Human-only | | | | | | Human + generated | | | | | |
|--------|----------------------------------|----------------|-----------------|------------------|------------------|-----------------|-----------------|-------------------|------------------|----------------|------------------|-----------------|-----------------|
| Change | Reference | 71.2 | 73.2 | 79.6 | 74.1 | 73.1 | 74.0 | 77.5 | 81.6 | 89.8 | 75.6 | 74.9 | 78.6 |
| | P1 · Criteria without guardrails | +3.2
(74.5) | -8.1*
(65.2) | -11.7*
(67.8) | -11.4*
(62.7) | +1.6
(74.7) | +5.3*
(79.3) | +4.1
(81.7) | -18.7*
(62.9) | -1.1
(88.7) | -12.1*
(63.4) | +1.2
(76.0) | +3.1*
(81.6) |
| | P2 · Criteria removed | +2.2
(73.5) | -0.2
(73.0) | -1.1
(78.4) | -3.0
(71.2) | -2.8*
(70.3) | -1.6
(72.4) | +4.0*
(81.6) | -0.6
(81.0) | +2.7
(92.5) | -0.6
(75.0) | -2.0*
(72.9) | -2.7*
(75.9) |
| | + Related work (retrieval) | +1.9
(73.1) | +5.0*
(78.3) | -0.4
(79.1) | -3.4
(70.8) | +0.7
(73.8) | +2.8
(76.7) | -3.6*
(73.9) | -1.0
(80.6) | +0.3
(90.1) | -4.1
(71.5) | +2.1
(77.0) | -0.3
(78.3) |
| | Reasoning effort = low | — | +2.5
(75.7) | -4.7*
(74.8) | -8.2*
(65.9) | -1.0
(72.1) | +2.8*
(76.7) | — | +2.5
(84.1) | -2.4
(87.4) | -1.0
(74.6) | +1.1
(76.0) | +2.8*
(81.4) |
| | No verdict aggregation | -0.7
(70.6) | +1.1
(74.3) | -2.4*
(77.2) | -2.8
(71.3) | -2.7*
(70.4) | -0.8
(73.2) | -2.2*
(75.4) | -0.7
(80.9) | +1.1
(90.9) | +1.5
(77.0) | +0.7
(75.6) | +0.5
(79.1) |
| | Idea as research plan | -1.6
(69.6) | -0.5
(72.7) | -3.3
(76.3) | +2.1
(76.2) | -2.5*
(70.6) | -1.1
(72.9) | -3.1*
(74.4) | -1.6
(80.0) | -0.6
(89.2) | +1.8
(77.4) | -2.5*
(72.3) | -3.6*
(74.9) |
| | | sonnet-4-5 | opus-4-5 | opus-4-6 | gpt-5.1 | gpt-5.2 | gpt-5.4 | sonnet-4-5 | opus-4-5 | opus-4-6 | gpt-5.1 | gpt-5.2 | gpt-5.4 |
| | | Judge model | | | | | | | | | | | |

* p < 0.05 (two-sided paired bootstrap)

* p < 0.05 (two-sided paired bootstrap)

(b) F1 on the **not-novel** class.

Figure 9: Per-class decomposition of the pointwise ablation results. Each cell reports the change relative to the reference configuration (top row), with the resulting absolute F1 in parentheses. Asterisks mark statistically significant changes.

claude-sonnet-4-5 tying on more than half of the pairs in certain configurations. Additionally, verdict aggregation substantially reduces ties.

Results without verdict aggregation. Figure 11 repeats the controlled study with aggregation turned off, calling pointwise judges once and pairwise judges once per presentation order. The reference configuration is the one from Figure 4 and still aggregates, so each cell reflects both the configuration change and the removal of aggregation. The trends reported in Section 5 persist. Pairwise scores drop once aggregation is removed, so gains shrink and losses deepen. For gpt-5.4 on Human+Generated, the gain from P3 drops from +15.6 to +12.3 points, while the loss from P4, already the

largest in the study, deepens from -37.0 to -46.8. Pointwise judges are less sensitive to aggregation: most scores drop by a few points, and some even rise slightly.

Prompt variants on research plans. Judges perform slightly worse when ideas are presented as research plans rather than abstracts (the *Idea as research plan* row of Figure 4b), mostly in the Human+Generated setting. To check whether the plan format also changes the effect of other design choices, we rerun the pairwise prompt variants P3–P5 (which produced the largest performance changes on free-form abstracts) with every idea rewritten as a plan. Figure 12 reports the results against the same reference as Figure 4b (the refer-

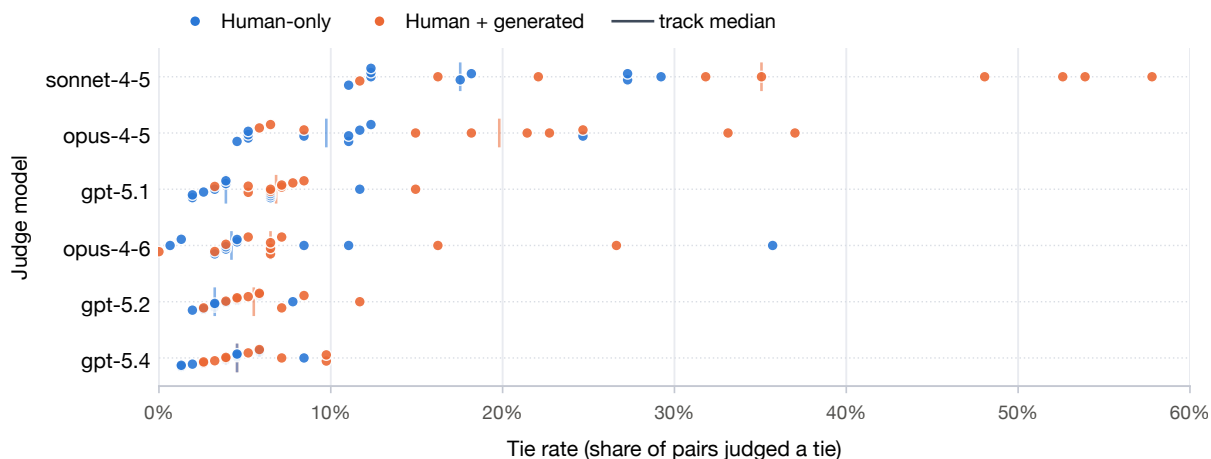


Figure 10: **Tie rates across controlled configurations.** Each point is one configuration of Figure 4b for a given judge (roughly 20 data points per judge), with the vertical bar marking the median per setting. Every judge ties more often in the Human+Generated setting than in the Human-Only one. Exact per-configuration values are in Table 6.

ence prompt on abstracts), so each cell reflects both the format change and the prompt change. Scores are generally a bit lower than with abstracts, but the overall trends persist: P3 still improves every judge on Human+Generated, P5 still degrades every judge in both settings, and P4 still produces the steepest drops, down to -55.8 points for gpt-5.4 on Human+Generated.

C.2.1 Different Ideation Models

To examine the effect of the idea generation model, we hold D_+ fixed and regenerate D_- with four ideation backbones (claude-sonnet-4-5, claude-opus-4-5, gpt-5.1, gpt-5.4), keeping the rest of the reference configuration (§4.1) unchanged. Figure 13 reports judge performance under each generator, alongside the Human-Only setup as a reference point, where D_- is human-authored.

Performance generally degrades once the negatives are LLM-generated, but pointwise and pairwise judges are not affected equally: pairwise accuracy suffers the most, falling below chance in several cases, whereas pointwise macro-F1 degrades more mildly and occasionally even improves. We also notice that the size of the drop does not necessarily track the strength of the generator: most pointwise judges rate ideas from claude-sonnet-4-5 as no less novel than those from the stronger claude-opus-4-5. This suggests that testing a novelty judge against human-authored ideas alone, or even against a single ideation system, is not enough: a judge that looks reliable against one type of data can invert against

another.

Retrieval across ideation backbones. Figure 4 and Figure 5 show that retrieval yields limited gains for novelty evaluation. We ask whether this still holds when D_- is generated by different backbones. The full grid of judges $\times$ generators $\times$ design choices grows quickly, so we re-run retrieval across four ideation backbones with a single judge from each model family (claude-opus-4-6 and gpt-5.4), and leave the complete cross-product to future work. Figure 14 reports the results. Consistent with our earlier findings, gains from retrieval are small and inconsistent, and in some cases retrieval significantly degrades performance.

C.3 When Retrieval Hurts

To understand why retrieval (counterintuitively) yields limited gains (Section 5.1), we inspect cases where it turns a correct verdict incorrect. We focus on one strong judge, gpt-5.4, on pointwise Human+Generated, where retrieval flips 22 of its correct verdicts to incorrect. Since each verdict aggregates three calls, a flip in the aggregated verdict could reflect sampling noise rather than an effect of retrieval. We therefore only consider ideas where the reference judge (no retrieval) is correct in all three calls and the retrieval judge is incorrect in all three. We sample five flips per class (5 from D_+ and 5 from D_- the reference judge classifies correctly). For each, we compare the judge’s reasoning with and without retrieval. We observe a few recurring patterns (see examples for each in Table 7).

| | | Human-only | | | | | | Human + generated | | | | | | |
|---------------------------------|--------------|----------------------------------|------------------|-----------------|-----------------|-----------------|-----------------|-------------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| Change (no verdict aggregation) | Judge prompt | Reference | 60.7 | 70.7 | 80.5 | 75.2 | 64.4 | 66.6 | 69.1 | 78.4 | 88.9 | 75.0 | 64.5 | 70.5 |
| | | P1 · Criteria without guardrails | +14.0*
(74.8) | +0.3
(70.9) | -6.3*
(74.2) | -6.2*
(69.0) | +6.4*
(70.9) | +6.2*
(72.8) | +8.8*
(77.9) | -6.8*
(71.6) | -0.3
(88.6) | -5.7*
(69.3) | +8.0*
(72.5) | +7.4*
(77.9) |
| | | P2 · Criteria removed | +9.1*
(69.8) | +5.9*
(76.5) | +0.4
(80.9) | -1.3
(73.9) | -8.9*
(55.6) | -6.5*
(60.1) | +7.4*
(76.5) | +3.9
(82.3) | +2.7
(91.6) | -3.3
(71.7) | -9.0*
(55.5) | -5.5*
(65.0) |
| | | + Related work (retrieval) | +4.4
(65.2) | +4.1
(74.8) | -1.0
(79.5) | -2.8
(72.4) | -0.8
(63.7) | +5.3
(71.9) | -2.6
(66.5) | -2.1
(76.3) | +0.0
(88.9) | -2.0
(73.0) | +5.3*
(69.8) | +3.1
(73.6) |
| | | Reasoning effort = low | — | +1.6
(72.3) | -5.3*
(75.2) | -5.0*
(70.2) | +0.7
(65.1) | +4.8*
(71.4) | — | +0.1
(78.6) | -3.4*
(85.4) | +0.5
(75.5) | +4.0*
(68.5) | +5.7*
(76.2) |
| | | Idea as research plan | +0.1
(60.8) | +2.2
(72.9) | -1.5
(79.1) | -1.0
(74.2) | -5.9*
(58.5) | -2.6
(64.0) | -5.9*
(63.2) | -4.8*
(73.7) | -1.2
(87.6) | +1.9
(76.9) | -4.0
(60.5) | -8.7*
(61.8) |
| | | | | sonnet-4-5 | opus-4-5 | opus-4-6 | gpt-5.1 | gpt-5.2 | gpt-5.4 | sonnet-4-5 | opus-4-5 | opus-4-6 | gpt-5.1 | gpt-5.2 |
| | | Judge model | | | | | | | | | | | | |

* p < 0.05 (two-sided paired bootstrap)

* p < 0.05 (two-sided paired bootstrap)

(a) Pointwise judging, macro-F1.

| | | Human-only | | | | | | Human + generated | | | | | | | |
|---|--------------|-------------------------------------|------------------|------------------|------------------|------------------|-----------------|-------------------|---------|------------------|------------------|-------------------|------------------|------------------|------------------|
| Change (no verdict aggregation) | Judge prompt | Reference | 74.0 | 79.2 | 82.5 | 83.8 | 83.8 | 83.1 | | 39.6 | 63.0 | 86.4 | 63.0 | 76.0 | 77.9 |
| | | P1 · Criteria without guardrails | -17.5*
(56.5) | -1.9
(77.3) | -5.8*
(76.6) | -3.9
(79.9) | -2.6
(81.2) | -3.9
(79.2) | | -27.3*
(12.3) | -22.7*
(40.3) | -9.1*
(77.3) | -18.2*
(44.8) | -17.5*
(58.4) | -17.5*
(60.4) |
| | | P2 · Criteria removed | -16.9*
(57.1) | -3.9
(75.3) | -3.9
(78.6) | -2.6
(81.2) | -1.9
(81.8) | -4.6
(78.6) | | -25.3*
(14.3) | -16.2*
(46.8) | -8.4*
(77.9) | -7.8*
(55.2) | -6.5*
(69.5) | -2.6
(75.3) |
| | | P3 · "Which was judged novel?" | -3.9
(70.1) | -5.8
(73.4) | -4.5
(77.9) | -0.6
(83.1) | -3.9
(79.9) | -4.6
(78.6) | | +16.9*
(56.5) | +27.9*
(90.9) | +13.6*
(100.0) | +7.8*
(70.8) | -1.3
(74.7) | +12.3*
(90.3) |
| | | P4 · "Which is novel?" | -7.2
(66.9) | -13.6*
(65.6) | -26.0*
(56.5) | -17.5*
(66.2) | -9.1*
(74.7) | -15.6*
(67.5) | | -5.8
(33.8) | -4.5
(58.4) | -13.6*
(72.7) | -13.6*
(49.4) | -5.2
(70.8) | -46.8*
(31.2) |
| | | P5 · "Which was judged more novel?" | -8.4*
(65.6) | -5.8
(73.4) | -5.8
(76.6) | -8.5*
(75.3) | -4.5
(79.2) | -2.6
(80.5) | | -13.6*
(26.0) | -3.9
(59.1) | -8.4*
(77.9) | -13.0*
(50.0) | -23.4*
(52.6) | -23.4*
(54.5) |
| | | + Related work (retrieval) | -13.0*
(61.0) | -4.5
(74.7) | +0.7
(83.1) | -4.5
(79.2) | -1.9
(81.8) | -1.3
(81.8) | | -2.6
(37.0) | +7.8*
(70.8) | -1.3
(85.1) | -5.8
(57.1) | -4.5
(71.4) | -7.8*
(70.1) |
| | | Reasoning effort = low | — | -1.3
(77.9) | -3.2
(79.2) | -1.3
(82.5) | -4.5
(79.2) | -9.7*
(73.4) | | — | -3.2
(59.7) | -3.2*
(83.1) | -11.7*
(51.3) | -9.1*
(66.9) | -8.4*
(69.5) |
| | | Idea as research plan | -12.3*
(61.7) | -7.1*
(72.1) | -2.6
(79.9) | -1.3
(82.5) | -0.6
(83.1) | -5.8*
(77.3) | | -21.4*
(18.2) | -14.3*
(48.7) | -9.7*
(76.6) | -5.2*
(57.8) | -4.5
(71.4) | -0.6
(77.3) |
| | | | | sonnet-4-5 | opus-4-5 | opus-4-6 | gpt-5.1 | gpt-5.2 | gpt-5.4 | sonnet-4-5 | opus-4-5 | opus-4-6 | gpt-5.1 | gpt-5.2 | gpt-5.4 |
| * p < 0.05 (two-sided paired bootstrap) | | | | | | | | | | | | | | | |
| Judge model | | | | | | | | | | | | | | | |

* p < 0.05 (two-sided paired bootstrap)

(b) Pairwise judging, strict-accuracy (ties count as failures).

Figure 11: **Results without verdict aggregation.** Cells report the change from the reference configuration, with the absolute value in parentheses. The reference row itself still aggregates. claude-sonnet-4-5 exposes no reasoning_effort parameter, hence the blank entry.

| | | Human-only | | | | | | Human + generated | | | | | | |
|--------------------------------|--------------|-------------------------------------|-----------------|------------------|------------------|----------------|----------------|-------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| Change (idea as research plan) | Judge prompt | Reference | 74.0 | 79.2 | 82.5 | 83.8 | 83.8 | 83.1 | 39.6 | 63.0 | 86.4 | 63.0 | 76.0 | 77.9 |
| | | Idea as research plan | -6.5*
(67.5) | -0.7
(78.6) | -1.9
(80.5) | +1.3
(85.1) | +2.6
(86.4) | +0.0
(83.1) | -11.7*
(27.9) | -4.5
(58.4) | -6.5*
(79.9) | -0.7
(62.3) | -2.6
(73.4) | +1.9
(79.9) |
| | | P3 · "Which was judged novel?" | +1.3
(75.3) | -4.5
(74.7) | -9.1*
(73.4) | -0.6
(83.1) | +0.0
(83.8) | -3.3
(79.9) | +10.4*
(50.0) | +20.1*
(83.1) | +9.7*
(96.1) | +9.7*
(72.7) | +3.9
(79.9) | +12.3*
(90.3) |
| | | P4 · "Which is novel?" | -5.2
(68.8) | -16.2*
(63.0) | -19.5*
(63.0) | -5.8
(77.9) | +0.0
(83.8) | -24.0*
(59.1) | +8.4
(48.0) | -13.0*
(50.0) | -35.7*
(50.6) | +0.6
(63.6) | +0.6
(76.6) | -55.8*
(22.1) |
| | | P5 · "Which was judged more novel?" | -2.6
(71.4) | -5.8
(73.4) | -7.2*
(75.3) | -3.2
(80.5) | -2.6
(81.2) | -4.6
(78.6) | -9.7*
(29.9) | -13.0*
(50.0) | -23.4*
(63.0) | -14.9*
(48.0) | -13.6*
(62.3) | -14.3*
(63.6) |
| | | | sonnet-4-5 | opus-4-5 | opus-4-6 | gpt-5.1 | gpt-5.2 | gpt-5.4 | sonnet-4-5 | opus-4-5 | opus-4-6 | gpt-5.1 | gpt-5.2 | gpt-5.4 |
| | | Judge model | | | | | | | | | | | | |

* p < 0.05 (two-sided paired bootstrap)

* p < 0.05 (two-sided paired bootstrap)

Figure 12: **Prompt variants with ideas as research plans,** pairwise strict-accuracy (ties count as failures). The first row applies the plan format alone, and rows P3–P5 combine it with the corresponding judge prompt. Scores are generally lower than with abstracts, but the effects of the prompt variants keep their direction.

| | | Pointwise — Macro-F1 | | | | | | Pairwise — Strict Accuracy | | | | | |
|------------------|------------|----------------------|----------|----------|---------|---------|---------|----------------------------|----------|----------|---------|---------|---------|
| Negatives source | Human | 60.7 | 70.7 | 80.5 | 75.2 | 64.4 | 66.6 | 74.0 | 79.2 | 82.5 | 83.8 | 83.8 | 83.1 |
| | sonnet-4-5 | 69.1 | 78.4 | 88.9 | 75.0 | 64.5 | 70.5 | 39.6 | 63.0 | 86.4 | 63.0 | 76.0 | 77.9 |
| | opus-4-5 | 66.2 | 72.7 | 84.4 | 65.1 | 61.6 | 70.0 | 18.2 | 45.5 | 68.8 | 47.4 | 55.8 | 57.8 |
| | gpt-5.1 | 54.2 | 59.5 | 73.1 | 50.4 | 57.9 | 64.5 | 3.2 | 24.0 | 49.4 | 23.4 | 28.6 | 34.4 |
| | gpt-5.4 | 53.2 | 45.3 | 60.5 | 35.3 | 52.5 | 66.5 | 4.5 | 20.8 | 38.3 | 11.7 | 19.5 | 18.2 |
| | | sonnet-4-5 | opus-4-5 | opus-4-6 | gpt-5.1 | gpt-5.2 | gpt-5.4 | sonnet-4-5 | opus-4-5 | opus-4-6 | gpt-5.1 | gpt-5.2 | gpt-5.4 |
| | | Judge model | | | | | | | | | | | |

Figure 13: **Effect of the lower-novelty ideas source.** Judge performance when the pool of high-novelty, human-authored ideas (D_+) is held fixed and the pool of lower-novelty ideas (D_-) is regenerated by each of four ideation backbones. Human-Only, where D_- holds human-authored ideas criticized for lacking novelty, is shown as a reference. Judge performance drops once the low-novelty side is LLM-generated, falling below chance level in some instances.

| | | Pointwise
Macro-F1 | | Pairwise
Strict Accuracy | |
|------------------|------------|---|----------------------|-----------------------------|----------------------|
| Negatives source | sonnet-4-5 | +0.3
(88.9→89.2) | +1.9
(70.5→72.5) | +0.0
(86.4→86.4) | -5.2
(77.9→72.7) |
| | opus-4-5 | -0.3
(84.4→84.1) | +1.8
(70.0→71.8) | +7.8*
(68.8→76.6) | +3.3
(57.8→61.0) |
| | gpt-5.1 | -2.3
(73.1→70.9) | -0.9
(64.5→63.6) | +2.6
(49.4→51.9) | +3.9
(34.4→38.3) |
| | gpt-5.4 | -4.6*
(60.5→55.9) | -7.2*
(66.5→59.3) | +1.9
(38.3→40.3) | +9.7*
(18.2→27.9) |
| | | opus-4-6 | gpt-5.4 | opus-4-6 | gpt-5.4 |
| | | Judge model | | | |
| | | Δ (without→with retrieval) | | | |
| | | * p < 0.05 (two-sided paired bootstrap) | | | |

Figure 14: **Retrieval across ideation backbones.** Effect of giving the judge retrieved related work, with D_+ held fixed and D_- regenerated by each of four ideation backbones (rows), for one judge per model family (columns). Retrieval yields small and inconsistent effects that do not agree in sign across the two protocols, and it leaves the hardest backbones far below chance.

A single close precursor decides the verdict.

This pattern underlies the D_+ flips (the reference judge correctly labels an idea novel and the retrieval judge labels it not novel). We observe that when one retrieved paper shares the idea’s headline framing, the judge treats the remaining contribution as incremental, e.g., a larger benchmark as “primarily a scale/resource extension”, or a new failure mode as “mainly a setting shift” (even when this part of the idea is praised by reviewers).

Component-wise matching. This pattern underlies the D_+ flips (the reference judge correctly la-

bels an idea novel and the retrieval judge labels it not novel). The retrieval judge maps each component of the idea to a different retrieved paper and concludes that the whole is “a fairly direct synthesis” (even when human reviewers explicitly describe the idea as a novel combination). Notably, without retrieval, the judge often recognizes the same components as known (“The core ingredients are not individually new”) yet still credits their combination as novel.

Retrieved work displaces prior knowledge.

This pattern underlies the D_- flips (the reference judge correctly labels a generated idea not novel and the retrieval judge labels it novel). Without retrieval, the judge dismisses the generated ideas as not novel based on its internal knowledge. With retrieval, it checks each idea against the five retrieved papers, finds none that covers it fully, and concludes it is novel (“does not appear in the provided related work”, “None of the cited papers ...”). Our retrieval prompt defines novelty “in relation to the provided related work” (Prompt 16), which likely encourages this over-reliance on retrieved papers.

ICLR area: applications to computer vision, audio, language, and other modalities

| | |
|--|---|
| 0
(validated novel) | <p>Despite advances in pretraining with extended context sizes, large language models (LLMs) still face challenges in effectively utilizing real-world long-context information, primarily due to insufficient long-context alignment caused by data quality issues, training inefficiencies, and the lack of well-designed optimization objectives. To address these limitations, we propose a framework named Short-to-Long Preference Optimization (SoLoPO), decoupling long-context preference optimization (PO) into two components: short-context PO and short-to-long reward alignment (SoLo-RA), supported by both theoretical and empirical evidence. Specifically, short-context PO leverages preference pairs sampled from short contexts to enhance the model’s contextual knowledge utilization ability. Meanwhile, SoLo-RA explicitly encourages reward score consistency for the responses when conditioned on both short and long contexts that contain identical task-relevant information. This facilitates transferring the model’s ability to handle short contexts into long-context scenarios. SoLoPO is compatible with mainstream preference optimization algorithms, while substantially improving the efficiency of data construction and training processes.</p> |
| 1
(weakly labeled lower-novelty) | <p>Multimodal understanding requires more than aligning representations across vision, audio, and language—it demands reasoning about temporal precedence and causal relationships between modalities. Existing approaches treat modalities symmetrically through attention mechanisms, failing to capture how information in one modality temporally influences or explains events in another. We introduce Causal Multimodal Transformers, a framework that explicitly models directional causal dependencies across asynchronous modality streams. Our architecture incorporates learned temporal offsets and causal masking patterns that respect the natural information flow between modalities, enabling the model to distinguish between coincidental co-occurrence and genuine causal influence. By decomposing cross-modal attention into temporally-ordered causal graphs, the framework learns which modality provides predictive information for events in others at different time scales. This causally-aware design enhances interpretability and enables counterfactual reasoning—answering questions like “what would the visual scene be if this sound had not occurred?” Applications span video understanding, audio-visual speech processing, and multimodal content generation where temporal causality is fundamental to meaning.</p> |
| Expert
selected: 0 | <p>The idea in 1 of causal/temporal dependence between modalities is old. Temporal offsets, causal masks with directional cross-modal attention etc. It’s all a generic shallow combination of these things. e.g., https://arxiv.org/abs/1906.00295 (>3K citations). The idea in 0 of decoupling long-context preference optimization into short and short-to-long seems clever and original, and a very specific method invention.</p> |

ICLR area: unsupervised, self-supervised, semi-supervised, and supervised representation learning

| | |
|--|---|
| 0
(weakly labeled lower-novelty) | <p>Representation learning paradigms—unsupervised, self-supervised, semi-supervised, and supervised—are typically treated as distinct methodologies requiring separate architectural designs and training protocols. We introduce Spectrum Learning, a unified framework that views supervision as a continuous spectrum rather than discrete categories. Our approach employs a meta-learned weighting mechanism that dynamically modulates the contribution of multiple learning objectives based on local supervision density in the feature space. By treating each data point as existing along a supervision gradient, the framework adaptively combines contrastive self-supervised losses, consistency regularization, pseudo-labeling, and supervised classification within a single coherent optimization. A novel supervision-aware attention module enables the model to identify which learning paradigm is most informative for different regions of the data manifold. This fluid integration allows seamless transitions as supervision availability changes, from fully unsupervised to fully supervised settings, without architectural modification. Spectrum Learning naturally handles mixed supervision scenarios common in practice, where different samples have varying levels of annotation quality and granularity, providing a principled approach to leveraging all available learning signals simultaneously.</p> |
| 1
(validated novel) | <p>Multi-view clustering integrates the consistency and complementarity of different views to achieve unsupervised data grouping. Existing multi-view clustering methods primarily confront two challenges: i) they generally perform feature extraction in the feature domain, which is sensitive to noise and may neglect cluster-specific information that is indistinguishable in the original space; ii) current dynamic fusion methods adopt static strategies to learn weights, lacking capability to adjust strategies adaptively under complex scenarios according to variations in data distribution and view quality. To address these issues, we propose a large language model assisted dynamic agent for multi-view clustering (LLM-DAMVC), a novel framework that recasts multi-view clustering as a dynamic decision-making problem orchestrated by a large language model. Specifically, each view is equipped with complementary agents dedicated to feature extraction. A dual-domain contrastive module is introduced to optimize feature consistency and enhance cluster separability in both the feature domain and frequency domain. Additionally, an LLM-assisted view fusion mechanism provides a flexible fusion weight learning strategy that can be adaptively applied to complex scenarios and significantly different views.</p> |
| Expert
selected: 1 | <p>1: multi view clustering, with view fusion with dynamic decisions controlled by an agent according to view quality, is interesting and not common (especially given the timeframe is 2025). 0: combining supervised losses, pseudo-labeling, consistency regularization, and unsupervised objectives, meta-learning sample weights... standard, generic and vague combination of standard things wrapped in purple prose “Spectrum Learning” which means nothing. https://arxiv.org/abs/1905.02249</p> |

Table 4: **Expert annotation examples.** Representative pairs from the blind annotation study on Human+Generated, alongside the rater’s written justification.

| Evaluator | Judge | k | F_1 | F_1^+ | F_1^- | Cost (USD) |
|---------------------------|-----------------|-----|--------------------------|--------------------------|--------------------------|------------|
| Human-Only | | | | | | |
| AI-Scientist | claude-opus-4-6 | 1 | 0.652 $\pm$ 0.055 | 0.743 $\pm$ 0.050 | 0.561 $\pm$ 0.079 | 77.91 |
| AI-Scientist | gpt-5.4 | 1 | 0.481 $\pm$ 0.056 | 0.293 $\pm$ 0.084 | 0.668 $\pm$ 0.053 | 27.41 |
| Idea Novelty Checker | claude-opus-4-6 | 1 | 0.560 $\pm$ 0.059 | 0.711 $\pm$ 0.050 | 0.408 $\pm$ 0.087 | 30.11 |
| Idea Novelty Checker | gpt-5.4 | 1 | 0.521 $\pm$ 0.057 | 0.701 $\pm$ 0.050 | 0.342 $\pm$ 0.088 | 29.98 |
| Prompted (no aggregation) | claude-opus-4-6 | 1 | 0.782 $\pm$ 0.047 | 0.792 $\pm$ 0.051 | 0.772 $\pm$ 0.054 | 4.56 |
| Prompted (no aggregation) | gpt-5.4 | 1 | 0.655 $\pm$ 0.055 | 0.578 $\pm$ 0.076 | 0.732 $\pm$ 0.051 | 5.23 |
| Prompted (reference) | claude-opus-4-6 | 3 | 0.806 $\pm$ 0.046 | 0.815 $\pm$ 0.048 | 0.796 $\pm$ 0.052 | 13.67 |
| Prompted (reference) | gpt-5.4 | 3 | 0.666 $\pm$ 0.053 | 0.592 $\pm$ 0.076 | 0.740 $\pm$ 0.050 | 15.68 |
| Prompted (low reasoning) | claude-opus-4-6 | 3 | 0.739 $\pm$ 0.050 | 0.729 $\pm$ 0.059 | 0.748 $\pm$ 0.053 | 2.43 |
| Prompted (low reasoning) | gpt-5.4 | 3 | 0.726 $\pm$ 0.051 | 0.685 $\pm$ 0.066 | 0.767 $\pm$ 0.049 | 4.66 |
| Prompted (retrieval) | claude-opus-4-6 | 3 | 0.802 $\pm$ 0.046 | 0.813 $\pm$ 0.048 | 0.792 $\pm$ 0.051 | 24.84 |
| Prompted (retrieval) | gpt-5.4 | 3 | 0.726 $\pm$ 0.052 | 0.685 $\pm$ 0.067 | 0.767 $\pm$ 0.050 | 19.36 |
| Human+Generated | | | | | | |
| AI-Scientist | claude-opus-4-6 | 1 | 0.772 $\pm$ 0.047 | 0.801 $\pm$ 0.047 | 0.743 $\pm$ 0.059 | 82.05 |
| AI-Scientist | gpt-5.4 | 1 | 0.481 $\pm$ 0.055 | 0.260 $\pm$ 0.085 | 0.702 $\pm$ 0.049 | 28.41 |
| Idea Novelty Checker | claude-opus-4-6 | 1 | 0.642 $\pm$ 0.055 | 0.728 $\pm$ 0.050 | 0.556 $\pm$ 0.077 | 30.66 |
| Idea Novelty Checker | gpt-5.4 | 1 | 0.526 $\pm$ 0.057 | 0.688 $\pm$ 0.051 | 0.365 $\pm$ 0.085 | 31.10 |
| Prompted (no aggregation) | claude-opus-4-6 | 1 | 0.902 $\pm$ 0.034 | 0.895 $\pm$ 0.039 | 0.909 $\pm$ 0.033 | 4.91 |
| Prompted (no aggregation) | gpt-5.4 | 1 | 0.718 $\pm$ 0.052 | 0.646 $\pm$ 0.073 | 0.791 $\pm$ 0.045 | 5.23 |
| Prompted (reference) | claude-opus-4-6 | 3 | 0.889 $\pm$ 0.036 | 0.879 $\pm$ 0.041 | 0.898 $\pm$ 0.035 | 14.74 |
| Prompted (reference) | gpt-5.4 | 3 | 0.705 $\pm$ 0.053 | 0.625 $\pm$ 0.075 | 0.786 $\pm$ 0.045 | 15.68 |
| Prompted (low reasoning) | claude-opus-4-6 | 3 | 0.854 $\pm$ 0.040 | 0.835 $\pm$ 0.049 | 0.874 $\pm$ 0.037 | 2.59 |
| Prompted (low reasoning) | gpt-5.4 | 3 | 0.766 $\pm$ 0.049 | 0.718 $\pm$ 0.064 | 0.814 $\pm$ 0.044 | 4.77 |
| Prompted (retrieval) | claude-opus-4-6 | 3 | 0.892 $\pm$ 0.035 | 0.883 $\pm$ 0.041 | 0.901 $\pm$ 0.034 | 25.19 |
| Prompted (retrieval) | gpt-5.4 | 3 | 0.725 $\pm$ 0.052 | 0.667 $\pm$ 0.069 | 0.783 $\pm$ 0.047 | 19.23 |

Table 5: **Pointwise results and cost** for every judge configuration, in the Human-Only (*top*) and Human+Generated (*bottom*) settings. k is the number of judge calls per decision, F_1 is macro-F1, and F_1^+ and F_1^- are the per-class scores on novel and non-novel ideas. Cost is the total USD spent judging the full set. Best macro-F1 per setting in **bold**. Subscripts are the half-width of the 95% bootstrap confidence interval over test instances, from 10,000 resamples. These numbers underlie Figures 6 and 7.

| Configuration | Human-Only | | | | | | Human+Generated | | | | | |
|-------------------------------------|------------|--------|--------|------|-----|-----|-----------------|--------|--------|------|------|-----|
| | son-4-5 | op-4-5 | op-4-6 | 5.1 | 5.2 | 5.4 | son-4-5 | op-4-5 | op-4-6 | 5.1 | 5.2 | 5.4 |
| Reference | 12.3 | 4.5 | 0.6 | 6.5 | 3.2 | 4.5 | 35.1 | 18.2 | 3.2 | 7.1 | 2.6 | 5.2 |
| P1 · Criteria without guardrails | 29.2 | 5.2 | 3.2 | 1.9 | 3.2 | 1.9 | 52.6 | 21.4 | 5.2 | 7.8 | 8.4 | 3.9 |
| P2 · Criteria removed | 27.3 | 11.0 | 4.5 | 2.6 | 2.6 | 3.9 | 57.8 | 24.7 | 7.1 | 6.5 | 3.9 | 2.6 |
| P3 · “Which was judged novel?” | 12.3 | 11.7 | 11.0 | 1.9 | 4.5 | 4.5 | 11.7 | 6.5 | 0.0 | 6.5 | 5.2 | 2.6 |
| P4 · “Which is novel?” | 12.3 | 24.7 | 35.7 | 6.5 | 5.8 | 4.5 | 16.2 | 37.0 | 26.6 | 7.1 | 4.5 | 9.7 |
| P5 · “Which was judged more novel?” | 11.0 | 11.0 | 8.4 | 3.9 | 1.9 | 1.3 | 22.1 | 22.7 | 16.2 | 8.4 | 5.8 | 5.8 |
| + Related work (retrieval) | 17.5 | 8.4 | 4.5 | 6.5 | 3.9 | 5.8 | 31.8 | 8.4 | 3.9 | 3.2 | 5.8 | 7.1 |
| Reasoning effort = low | | 5.2 | 3.9 | 3.2 | 3.2 | 5.8 | | 5.8 | 6.5 | 5.2 | 4.5 | 3.2 |
| No verdict aggregation | 27.3 | 12.3 | 1.3 | 11.7 | 7.8 | 8.4 | 53.9 | 33.1 | 6.5 | 14.9 | 11.7 | 9.7 |
| Idea as research plan | 18.2 | 5.2 | 3.9 | 3.9 | 3.2 | 1.3 | 48.1 | 14.9 | 6.5 | 5.2 | 7.1 | 3.9 |
| Mean | 18.6 | 9.9 | 7.7 | 4.9 | 4.0 | 4.2 | 36.6 | 19.3 | 8.2 | 7.2 | 6.0 | 5.4 |

Table 6: **Tie rates (%) across controlled configurations**, the numbers behind Figure 10. Judges are abbreviated: son-4-5 is claude-sonnet-4-5, op-4-5 is claude-opus-4-5, op-4-6 is claude-opus-4-6, 5.1 is gpt-5.1, 5.2 is gpt-5.2, 5.4 is gpt-5.4. Judges are substantially more uncertain on the Human+Generated set.

Idea (D_+): "...Multi-Agent System (MAS) powered by Visual Language Models ... suffers from a novel failure term, multi-agent visual hallucination snowballing, where hallucinations are seeded in a single agent and amplified by following ones... we propose ViF, a lightweight, model-agnostic mitigation paradigm..."

Reviewers: "The authors introduce a new task setting for MAS hallucination and create a multi-turn evaluation metric which could be beneficial for future researchers."

Without retrieval: "...although several ingredients have precedents, the problem setting plus the specific diagnosis-and-mitigation package appears substantively original..." → novel ✓

Retrieved: *Investigating and Mitigating the Multimodal Hallucination Snowballing in Large Vision-Language Models* (2024)...

With retrieval: "...although the MAS-specific framing is somewhat new, the main concept, analysis direction, and mitigation ingredients are all strongly anticipated by the provided related work..." → not novel ✗

Idea (D_+): "...we revisit Sobolev integral probability metrics (IPM) on trees to obtain a practical generalization of TSW..."

Reviewers: "The idea of combining tree-sliced Wasserstein with the Sobolev IPM to handle the general p -Wasserstein distance is novel"

Without retrieval: "The core ingredients are not individually new... However, the specific contribution described here appears to be a genuinely new synthesis" → novel ✓

Retrieved: [1] *Sobolev Transport: A Scalable Metric for Probability Measures with Graph Metrics* (2022); ... ; [4] *Tree-Sliced Variants of Wasserstein Distances* (2019)

With retrieval: "a direct combination of existing ingredients. [1] already introduces the tree-sliced paradigm... [4] already introduces a closed-form Sobolev transport / Sobolev IPM on graph metric spaces.... Therefore, defining "Tree-Sliced Sobolev" by averaging a Sobolev/tree discrepancy over random trees is a fairly direct specialization/composition of [1] and [4], rather than a new framework." → not novel ✗

Idea (D_- , generated): "We introduce ChemoSense, a multimodal framework that learns computational representations of taste and smell...by leveraging their correlations with vision, language, audio, and molecular structure"

Retrieved: *SMILES to Smell* (2021); *Recipe1M+* (2021); *DeepNose* (2018); *VirtualTaste* (2021); a recipe-retrieval model (2021)

Without retrieval: "this looks more like a broad composition of known components than a clearly novel framework, task, or resource." → not novel ✓

With retrieval: "...the idea is not novel at the level of basic method motifs. However the proposal does appear novel in its overall problem formulation and synthesis relative to the provided related work. None of the cited papers explicitly ... align chemical compounds with culinary multimodal data ... in one shared embedding space..." → novel ✗

Table 7: **Examples of retrieval hurting gpt-5.4** (Human+Generated, pointwise). In the first row, a single close precursor outweighs the contribution the reviewers valued. In the second, the judge matches each component to a different retrieved paper and dismisses the combination the reviewers called novel. In the third, the retrieved papers override the judge’s own recognition that the idea is a broad composition of known components: since no single retrieved paper covers the idea in full, the judge concludes it is novel.